

Collision Geometry Is Linear

Diophantine Certificates for Rational Signal Machines

Event-sparse chambers, canonical quadratic and quartic certificates,
and two routes to complete chronology

AI-assisted research report prepared for Vladimir Reshetnikov from two manuscripts
(batch 78 of ProveIt’s incoming reports, manuscripts 07 and 11), 2 October 2026

Abstract

A rational signal machine computes by the collisions of finitely many kinds of particles moving at fixed rational speeds on a line. This report prints two independent manuscripts that prove one theorem by two routes: once a complete finite collision history is fixed, its realizations are exactly the natural zeros of a sum of squares of integer affine residuals, of degree two, with exactly one witness tuple for each realization. In both, typed endpoint conditions on maximal adjacency intervals exclude unrecorded earlier collisions, omitted simultaneous inputs and omitted simultaneous sites, and causal depth controls the denominators of the event coordinates.

Part I (source 07) eliminates every event coordinate. The realizing initial-gap vectors form a relatively open rational polyhedral cone cut out by at most $3(n - 1) + (3m + 1)E$ linear conditions, and the only witnesses are the slacks of the strict rows. The chamber geometry yields small integral realizations, an NP upper bound for bounded synthesis, a robustness–codimension criterion, a canonical first-hit quartic for a fixed event budget, exact quasi-polynomial seed counts, an exactly solved first-collision tournament and a four-speed accumulating clock with exponentially growing denominators. Part II (source 11) keeps batch times and event positions as natural witnesses on a fixed rational grid, adds halting semantics, proves that the witness matrix has full column rank (so the certificate is strictly convex in its witnesses), gives linear witness and residual ledgers and a full-layer reference construction, and bounds the canonical witnesses by $60^K \max(1, M_0)$ for the four-speed interface of Durand-Lose’s universal machine.

Neither Part compiles unbounded halting into one fixed-arity polynomial, transcribes the published universal machine, or improves the repository’s 75- and 87-operation bounds. Part I’s convex-quadratic barrier is a special case of Part XI of the collection’s report *canonical-diophantine-certificates*. The report is AI-assisted and unrefereed; nothing in it is formalized in Lean or Rocq.

Status. Conventional proofs and finite exact-arithmetic checks; not kernel-checked. Section 1 compares the two routes and states what is not claimed; Appendix E gives the provenance.

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1 Scope, the two routes, and conventions

[write] This section was written when the two manuscripts were merged. Part I and Part II print the two sources; text there without a [write] marker is the source’s own.

1.1 The two sources

This report is built from two AI-assisted research manuscripts that arrived in one delivery as batch 78 of ProveIt’s incoming reports (manuscripts 07 and 11; Appendix E gives the full provenance). They are called *source 07* and *source 11* after their batch-78 manuscript numbers, which are also the file prefixes of their shipped programs and data. Neither cites the other, and a text-overlap check at placement found almost no shared wording (at most 0.5 per cent of either text’s eight-word sequences occur in the other). They nevertheless prove the same theorem, by the same mechanism, in two different witness formats (Section 1.2).

- **Part I (source 07, the base), *Collision geometry is linear*.** Arbitrary rational speeds, of either sign. Every event time and position is eliminated as a rational linear form in the initial gaps, so a complete chronological skeleton becomes a homogeneous rational chamber (Theorem 5.3) and a convex quadratic with unique slack witnesses (Theorem 6.1). The chamber geometry then gives denominator and height bounds, small integral realizations, an NP upper bound, robustness and codimension, finite-horizon semilinearity, a canonical first-hit quartic, exact Ehrhart counts, two exact examples and a convex-quadratic barrier.
- **Part II (source 11), *Direct Diophantine certificates*.** Speeds and positions normalized to natural numbers by an external Galilean change of coordinates. Batch times and event positions stay in the certificate as natural witnesses on the grid $B = D^K$, and the certificate has prefix and halting versions (Theorem 18.3). Its additions are the halting semantics with terminal rays, full column rank of the witness matrix and hence strict convexity in the witnesses (Lemma 22.1), an explicit witness and residual ledger, a full-layer reference construction with exact counts (Section 23), the witness-height bound for Durand-Lose’s four-speed interface (Theorem 25.2) and a positive-witness translation.

Source 07 is the base because it assumes less (signed rational speeds, no normalization interface) and proves more about the chamber; source 11 is printed in full as the second route together with its additions. Each Part keeps its source’s definitions, letters and numbering of statements. The common theorem is stated in both Parts, because its two statements have different inputs and witnesses; where a lemma of Part II coincides with one of Part I, a [write] note at the lemma says so, and source 11’s proof is kept as a marked second route.

1.2 One theorem by two routes

The common theorem. Fix a rational signal machine, an initial label word and a finite chronological collision history (a *skeleton* in Part I, a *schema* in Part II). There is an effective sum of squares of integer affine residuals, in the input coordinates and natural witnesses, whose natural zeros are exactly the realizations of that history, with exactly one witness tuple for each realization; its degree is at most two. This is Theorems 5.3 and 6.1 of Part I and Theorem 18.3 of Part II (for prefix semantics).

The common mechanism. Both sources record each pair of adjacent signals once, over the

maximal interval during which it stays adjacent, and impose a *typed* condition at each end of that interval: equality exactly when both segments are born (or die) at the same collision there, and strict positivity otherwise, including for a pair that survives the last listed batch. Affineness of the gap then excludes every interior meeting (Part I, Lemma 5.2; Part II, Lemma 21.1), and the strict conditions at a batch boundary make the declared blocks exactly the collision sites (the soundness part of the proof of Theorem 5.3; Lemma 21.2). The two sources then prove the same auxiliary statements: the lifetime count $(n - 1) + \sum_e (q_e + 1)$ ((5.2) and (18.2), where Part II writes $|\text{out}(e)|$ for q_e), causal-depth denominators dividing D^h (Theorem 7.1, Lemma 19.1), and semilinearity of the inputs accepted under a fixed event bound (Theorem 10.1, Proposition 26.1). Table 1 compares the two formats.

Table 1: The two routes compared.

	Part I (source 07)	Part II (source 11)
Speeds	arbitrary rationals, of either sign; time rescaled by $\tau = t/L$ to make them integers (Section 3)	natural numbers, after the external change of coordinates (19.1) (a Galilean shift and two scalings)
Input	the initial gap vector g , translation removed; real in the chamber theorem, positive integers in the certificate	the absolute initial positions $\mathbf{p}$, natural numbers in increasing order
Event times and positions	eliminated: rational linear forms in g , computed data (Section 4)	witnesses: natural T_i and X_e , scaled to the grid $B = D^K$ (18.1)
Witnesses	one slack $z_j = G_j g - 1$ per retained strict row; s in all	batch times T_i , time slacks d_i , event positions X_e , input-order slacks u_j , endpoint slacks $h^\pm$; $W \leq E + 2K + 2L + n_-$
Size	at most $3(n - 1) + (3m + 1)E$ raw conditions (5.3); a row may be dense in g	$R \leq I + K + 2L + n_- \leq (r + 2s + 3)E + 3n_-$ residuals, each of bounded width for a fixed machine (18.3)–(18.4)
Denominators	D^h at an event of causal depth h , for the coefficients of its forms	D^h for coordinates of depth h ; the compiler clears with D^K , a multiple of D^H
Convexity	convex, Hessian $2A^\top A \succeq 0$; unique slack by (6.3)	strictly convex in the witnesses for fixed input: full column rank over $\mathbb{R}$, even for unrealizable schemas (Lemma 22.1)
Semantics	prefixes: exactly the first B batches; nothing after them	prefixes, and halting realizations certified by terminal rays
Heights	row coefficients at most $2d(1+V)(DK)^h$ (Proposition 7.3); a realizing seed of size at most $d^{d/2}A^d$ (Theorem 8.1)	every canonical input and witness at most $60^K \max(1, M_0)$ for speeds in $\{0, 2, 3, 4\}$ (Theorem 25.2)
Shuttle	four-speed clock, speeds $0, 2, -1, -2$; denominator exactly $2 \cdot 3^k$ at event $2k$ for the seed $(1, 2)$; Zeno time $S = u + v$ (Section 12)	bounded shuttle, speeds $3, 2, 4, 0$; denominator exactly 3^{k-1} at the k th left event; Zeno time 2 (Proposition 25.3)
Missed simultaneity	two independent sites, word P, N, P, N , gaps $(2, 5, 2)$ (Section 11.3)	hidden collision at the final cut, word d, a, d, a at $(0, 4, 10, 14)$ (Section 24.2)

	Part I (source 07)	Part II (source 11)
Only here	chamber geometry, small realizations, real = rational = integer realizability, NP bound, robustness and codimension, first-hit quartic, Ehrhart counts, the tournament, the convex-quadratic barrier	halting semantics, strict convexity, the full-layer construction and its exact counts, the four-speed height bound, positive witnesses

Normalizations. The two definitions of D agree. Part I multiplies all speeds by an integer L and leaves their signs; Part II subtracts a common speed c and multiplies by an integer m . A common shift does not change speed differences, so with $m = L$ both take D to be the least common multiple of the same nonzero integer differences. Part II needs nonnegative speeds and an origin because its inputs are natural absolute positions; Part I needs neither because its input is the gap vector. Part I’s integer seeds come from homogeneity (scaling a rational chamber point); Part II fixes the scale outside the witness tuple and warns that a scale witness or variable denominators would not preserve its canonical quadratic claim without a new argument (Section 19.1).

Two shuttles, one mechanism. Part I’s clock and Part II’s shuttle are both a particle reflected between two converging walls, each round trip shrinking the remaining time to the accumulation by a factor three, so that the reduced denominators grow exponentially in the number of events while their bit lengths grow linearly. They are different machines: Part I’s has a stationary left wall and a right wall of speed -1 , and its every finite prefix has the whole positive quadrant as chamber (robust chronology with exponentially growing precision); Part II’s lives entirely in $0 < t < 2$, $0 < x < 6$ and is stated for its natural-speed interface. Neither source gives the accumulation point a continuation. Source 07 credits the accumulation phenomenon to Becker et al. [6].

Two missed simultaneities, one failure mode. In both examples a proposed batch omits a second, spatially separate collision that happens at the same time, and in both the omitted pair is an unchanged adjacent pair whose gap reaches zero exactly at the last listed boundary. The strict upper endpoint of a surviving pair rejects it: in Part I through the chamber row that must be positive there, in Part II through the residual $G(T_1) - h^+ - 1 = -h^+ - 1$. Part II adds the observation that weaker conditions (two nonnegative endpoint gaps with positive sum) would accept the false schema.

Two meanings of “sparse”. In Part I “event-sparse” counts geometric conditions: the bound (5.3) is linear in n and E , but a propagated intercept may depend on every initial gap, so the rows can be dense and storing them can take $O(dR_\sigma)$ coefficients (the paragraph after Theorem 5.4). In Part II the sparse certificate also has residuals of bounded width, because event coordinates are witnesses; its coefficients then contain the grid constant $B = D^K$. *Neither* size statement is a constant arithmetic-operation count for an unbounded computation, as both sources say.

1.3 Notions and letters

Each Part keeps its source’s letters. Many letters carry different meanings in the two Parts; Table 2 lists every letter used in both, so that a statement of one Part is never read with the other Part’s meaning. Inside a Part the meaning is always the one in that Part’s column. No symbol was renamed and no normalization was changed: source 11’s macro for the natural numbers is called `\NN` in this file and still prints $\mathbb{N}$, and Part I prints $\mathbb{N}_0$; both denote $\{0, 1, 2, \dots\}$.

Table 2: Letters used in both Parts.

Letter	Part I (source 07)	Part II (source 11)
$\mathbb{N}_0, \mathbb{N}$	$\mathbb{N}_0 = \{0, 1, 2, \dots\}$	$\mathbb{N} = \{0, 1, 2, \dots\}$ (the same set)
A	coefficient bound (7.2); a matrix in proofs	witness coefficient matrix (Lemma 22.1); birth-position expression A_σ ; start coordinates $A_{i,j}$
B	number of batches; a nonsingular matrix in the proof of Theorem 8.1; a label of the tournament	the grid modulus $B = D^K$ (18.1), <i>not</i> a number of batches; a label of the shuttle
C	the chamber $\mathcal{C}$; the count $C(H)$; a label	matrix of input coefficients; end coordinates $C_{i,j}$
D	lcm of nonzero speed differences (7.1)	the same lcm (18.1) (Section 1.2)
d	$d = n - 1$	time slacks d_i ; a label
E	number of collision sites; $\mathbf{E}$ the equality matrix	number of events (a site of Part I)
G	$\mathbf{G}$ the strict-row matrix	the gap function $G_{\sigma\tau}(T)$ (20.5)
g	the initial gap vector	number of blocks g_i of a slab (Section 23); a gap in proofs
H, h	h causal depth; H the side of a counting box	h a depth bound, H the maximal depth; $h, h^\pm$ endpoint slacks
K	$K = 2 + 4V$ (Lemma 7.2); number of first-hit chambers	number of batches, <i>not</i> a number of chambers
L	integer clearing the speeds; a label of the clock	number of recorded lifetimes; a label of the shuttle
M	the label set	largest position M_j ; $\mathcal{M}$ the machine
m	number of distinct speeds	the speed multiplier of (19.1); number of generators of a linear set
n	number of initial signals	the same; $n_- = \max(n - 1, 0)$
p, q	p_e, q_e incoming and outgoing counts at site e	initial positions p_j ; the position scale q
R	raw row count R_σ ; a label	residual count; a label
r, s	numbers of retained equality and strict rows	largest incoming arity and largest fanout; residuals r_a
S	the accumulation time $u + v$ of the clock	a label; $\mathcal{S}$ the schema
T	the site budget	scaled batch times T_i
u, v	initial gaps of the examples; $v(a)$ a speed	input-order slacks u_j ; $u(\ell)$ an original speed, $v(\ell)$ a normalized speed
V	$\max(1, \max v(a))$	the maximal speed (remark after Theorem 25.2)
W	$W = 3u + 2v$ of the clock	number of witnesses
z, y	z the slack witnesses	$\mathbf{y}$ the witnesses; $\mathbf{z}$ a test vector

Three words need the same care.

- **Skeleton** (Part I) and **schema** (Part II) carry the same information: an ordered list of batches, each a set of disjoint contiguous blocks of the live list. A skeleton lists only the collision blocks (at least two members); a schema partitions the whole live list and marks singletons as persisting signals. A *site* of Part I is an *event* of Part II.
- **Realization.** Part I's realization and Part II's prefix realization are the same notion: exactly the listed batches as the first batches, nothing asserted afterwards. Part II's *halting* realization

additionally excludes every future collision. The “halting relation” of Part I’s Section 13 is the undecidable halting of a universal machine, a different thing.

- **Convex.** Part I’s certificate is convex (positive semidefinite Hessian) jointly in seed and slacks; Part II’s is *strictly* convex in the witnesses with the input held fixed, and Part II states that joint strict convexity is neither needed nor generally true.

1.4 Relation to the repository and to the literature

Neither source cites the related material below; this subsection supplies the cross-references (pins and dates in Appendix E). Reports of the collection are cited by label name, since their labels do not resolve in this file.

- **Report *canonical-diophantine-certificates*, Part XI** [11], “Order-free self-assembly and a positivity-restricted degree threshold”. Its Lemma `cdc:of:lem:affinezero`, Theorem `cdc:of:thm:cubicsemilinear` and Theorem `cdc:of:thm:classification` ($D_2^+(k) = D_3^+(k) = \text{SL}(k)$) prove, for polynomials of degree at most three that are nonnegative only on the real nonnegative orthant, that the full natural zero set is a finite union of natural solution sets of rational affine systems, hence semilinear and effectively so. Part I’s Theorem 13.1 and Corollary 13.2 are the globally nonnegative quadratic special case; they are printed with source 07’s direct proof, as a second route, and credited in Remark 13.3. No novelty is claimed for them. The same Part’s Theorem `cdc:of:thm:singlefoldsl` (a single-fold orthant-nonnegative quadratic for every semilinear set) bears on Part I’s first-hit quartic; see Remark 10.3. The bounded-search argument at the end of Part I’s Section 13 is the standard one of Proposition `cdc:bd:prop:nobound` there, for a different statement (site counts of accepting runs).
- **Report *liveness-beyond-halting*** [12]. Its Part V prints a question of its source 11, “Optimal degree under stronger geometric restrictions”, which asks for the minimal degree and dimension of a witness-free, binary-branching local polynomial that is globally nonnegative on all real inputs, and remarks that the globally nonnegative quadratic case “has a much more rigid affine geometry”. Part I’s Theorem 13.1 (and, more generally, Part XI of [11]) states that geometry: such a quadratic has a rational affine zero set. This report does not settle that degree question.
- **The Hilbert-tenth-problem programme has already reviewed both manuscripts.** The review of the delivery of 2 October 2026 [13], with its detailed signal-machine part [14], replayed both suites from the original archives, read both articles in full and reports “PASS for both reports’ stated finite-schema, exact-domain theorems and generated examples”, with no mandatory correction. It also proves and checks two reductions that neither source claims: deleting source 11’s identically zero same-birth rows and its same-death rows (differences of retained flight rows), and erasing the initial-order slacks when $K > 0$ or the certificate is halting, which takes the worked annihilation from six residuals and five witnesses to four and four (a bijection of natural zero sets, not of nonnegative rational ones, with a rational counterexample); and deleting redundant strict rows, with computed slacks, from the five exported packets of source 07. Those reductions are the review’s, are not re-proved here, and do not change any theorem of either Part. This report’s intake is therefore not a first review; it adds the merge, the cross-references and the credit of Remark 13.3.
- **The programme’s audits of finite-history models.** Source 07 cites the reversible finite-

history audit [2] for its distinction between cheap local relations and a complete computation certificate; that file is unchanged from source 07’s pin to the write. A sibling audit [15] notes that “finite particles can in principle store unbounded values in their separation” and that “a new construction would have to establish that interface”. Durand-Lose’s halting universal signal machine [3] uses a finite initial configuration and a halting event (Part II, Section 17), so signal machines are a candidate for such an interface, and both Parts compile its finite collision histories in principle; but neither transcribes its universal rule table or its input loader (Part I, Section 14; Part II, Question 1), so this report does not establish that interface.

- **The repository’s universal bounds.** The project README [1] still states the established bounds of 75 operations for a complete certificate and 87 for one universal polynomial. Neither Part improves or bears on them; both say so.
- **Other reports of the category.** No result of *probabilistic-quantum-and-continuous-computation* or *group-theoretic-substrates* is re-proved here. Apart from the programme’s review described above, nothing in the repository treated signal machines before this report.

1.5 Status and what is not claimed

The report is AI-assisted and unrefereed, and **nothing in it is formalized in Lean or Rocq**. Both sources give conventional proofs together with finite exact-arithmetic checks that are regression evidence, not proofs. Every limitation stated by either source is kept in its Part; collected here, neither Part claims

- historical priority, or exhaustive novelty, for its constructions (Part I, Section 2; Part II, Section 17); Part II adds that a general linearizability of signal-machine evolution is already in Durand-Lose’s linear real-register simulation [4];
- one fixed-arity polynomial for unbounded halting, a single-fold or finite-fold Diophantine representation of c.e. sets, or a fixed-universal arithmetic-operation bound: the skeleton or schema, the machine and the input length are compiler data, and the row and residual counts are not operation counts;
- that the published universal machine (13 meta-signals, 21 non-blank rules) or its input loader has been transcribed, executed or certified;
- any semantics at or after an accumulation point; the clock and the shuttle are tests of finite prefixes, and the clock is not a claim to have discovered accumulation;
- NP-hardness (Corollary 8.3 is an upper bound), an efficient enumeration of first-hit chambers or of their volumes, or an optimized exhaustive enumerator in the code;
- that the denominator-growth examples are lower bounds for arbitrary Diophantine representations: Part II’s is a lower bound for common-grid encodings;
- that integer scaling of a rational seed preserves a prescribed physical length or deadline;
- that the finite checks are exhaustive, or that the shared simulator of Part II’s two compilers is independently verified;
- any Lean or Rocq verification; neither source’s code is a verified compiler.

The report itself claims no novelty for Part I’s convex-quadratic barrier (Remark 13.3) and none for the reductions of the programme’s review.

1.6 Research questions

Both agendas are printed in full: the nine research directions of Part I (Section 15) and the five questions of Part II (Section 27.1), each in its Part and with its source’s wording. Four of Part II’s questions coincide with, or overlap, directions of Part I; Table 3 pairs them, so that the two formulations are read together. None of them is answered by the other source, by the repository, or by the programme’s review; the one partial overlap is that Part II’s Question 3 begins with causal-depth clearing, which both sources already prove (a [write] note there re-scopes it).

Table 3: Concordance of the research questions.

Part I (source 07)	Part II (source 11)	Relation
15.2 An audited universal signal-machine frontend	Q1 Instantiate the universal table and input map	the same task
15.1 Formalizing the chronology theorem	Q5 Formalize geometric completeness	the same target (endpoint lemma and sweep invariant; typed endpoints and atomic replacement)
15.3 Sharper denominator invariants; 15.4 From few rows to small arithmetic circuits	Q3 Sharpen the denominator and cost ledger	overlapping: prime-valuation denominators, and a literal gate ledger
15.5 Avoiding exhaustive first-hit enumeration	Q4 Compress histories without losing canonicity	overlapping: both separate bounded compression from fixed-arity unbounded compression
15.6, 15.7, 15.8, 15.9	—	Part I only
—	Q2 Exploit conservative bounded populations carefully	Part II only

Part I

Collision geometry is linear (source 07)

[write] Part I prints batch-78 manuscript 07, *Collision Geometry Is Linear: Event-Sparse Quadratic Diophantine Certificates for Rational Signal Machines*, “research manuscript prepared with ChatGPT for Vladimir Reshetnikov’s ProveIt research program”, dated 2 October 2026. Its labels carry the prefix `smc:cg::`; its sections are Sections 2–16 here and its two appendices are Appendices A and B. Its abstract and keywords follow.

Abstract (source 07). A rational signal machine computes by the collisions of finitely many types of particles moving at fixed rational velocities on a line. Its continuous geometry seems to require nonlinear arithmetic certificates. We show that, once a complete chronological collision skeleton is fixed, the geometry is instead governed by homogeneous rational linear equations and strict linear inequalities in the initial gaps. A sweep construction using maximal adjacency intervals proves that these conditions exclude both unrecorded earlier events and omitted simultaneous collisions. For n initial signals, m distinct velocities, and E collision sites, at most $3(n - 1) + (3m + 1)E$ raw linear conditions suffice.

After positive denominator clearing, one globally nonnegative convex quadratic polynomial certifies the skeleton over the nonnegative integers. Every accepted integer seed has exactly

one slack-witness tuple. A finite disjoint first-hit family then yields one quartic with unique selector and slack witnesses. We also prove causal-depth denominator and coefficient bounds, a small integral realization theorem, a robustness–codimension criterion, and a finite-horizon quasi-polynomial counting theorem. Explicit examples include a first-collision tournament with an exact counting formula and inverse, and a four-speed accumulating clock whose denominator size grows exponentially in causal depth. Finally, we prove that globally nonnegative quadratic certificates cannot be compressed into a single fixed-arity universal halting representation without leaving their decidable linear zero-set class. ([write] This last result is a special case of Part XI of the collection’s report *canonical-diophantine-certificates* [11]; see Remark 13.3.)

These are manuscript-level results with executable exact-arithmetic checks, not proof-assistant-certified theorems. The generic compiler and regression suite accompany the article. No priority claim, resolution of finite-fold MRDP, or improvement to an existing fixed-universal operation bound is made.

Keywords. Diophantine representation; signal machines; collision chronology; rational polyhedral cones; Presburger arithmetic; Ehrhart quasi-polynomials; Zeno accumulation; exact computation.

2 The question and the contribution

2.1 A concrete gap between universality and arithmetic

The MRDP theorem guarantees that a computably enumerable acceptance relation has a Diophantine representation. That existence theorem does not itself give a transparent arithmetic description of a particular computational substrate. A useful substrate-specific compiler should explain what its witnesses mean, why they cannot describe a counterfeit computation, and which resource bound is actually being measured. The distinction is especially important when the substrate has a geometric, asynchronous, or apparently continuous dynamics. For background on the classical theorem and a machine-based formalization, see Matiyasevich [9] and Bayer et al. [10].

The **ProveIt** Hilbert-tenth-problem material motivates precisely this separation. Its reversible finite-history audit distinguishes cheap local relations from a complete finite-input computation certificate, and points to additional obligations concerning wiring, initialization, boundaries, and halting observations [1, 2]. This article develops an analogous complete *geometric frontend* for a different substrate: one-dimensional rational signal machines. It does not attempt another operation-count reduction for the repository’s universal polynomial.

[write] The repository has since reviewed this manuscript, and the related manuscript printed as Part II, in its Hilbert-tenth-problem research tree (Section 1.4). The reversible-history audit cited here is unchanged from this source’s pin to the write.

Signal machines are not merely collections of unrelated finite circuits. Durand-Lose gives a halting universal construction with 13 signal types and 21 collision rules, obtained through cyclic tag systems [3]. This established universality is the reason to study the substrate. We do not reproduce or independently certify that universal machine’s full rule table and input loader here. Instead we prove a generic finite-history compiler whose interface applies to the rational fixed-speed model.

The central question addressed is the following.

Can an explicitly proposed finite collision history be represented arithmetically without introducing arbitrary event coordinates, without overlooking an earlier collision, and without splitting a genuinely simultaneous event into a false sequence?

Our answer is affirmative. In fact, the required geometric residuals can be linear rather than quadratic, and their sum of squares can therefore be quadratic rather than quartic. The essential work is not squaring equations; it is showing that a small set of order constraints certifies the *complete* chronology.

2.2 Results and their logical scope

For a fixed initial label word and a fixed discrete skeleton σ , we construct integer matrices $\mathbf{E}_\sigma, \mathbf{G}_\sigma$ such that the real initial-gap vectors realizing that skeleton are exactly

$$\mathcal{C}_\sigma = \{g \in \mathbb{R}^{n-1} : \mathbf{E}_\sigma g = 0, \mathbf{G}_\sigma g > 0\}. \quad (2.1)$$

Initial gap positivity is included among the strict rows. Every event time and position is a rational linear form in g . Section 5 proves the event-sparse bound

$$R_\sigma \leq 3(n-1) + (3m+1)E \quad (2.2)$$

for the number of raw linear residuals and inequalities.

The corresponding natural-number certificate is

$$P_\sigma(g, z) = \sum_i (\mathbf{E}_{\sigma,i} g)^2 + \sum_j (\mathbf{G}_{\sigma,j} g - 1 - z_j)^2. \quad (2.3)$$

Its degree is at most two, and its entire auxiliary tuple is forced to be $z_j = \mathbf{G}_{\sigma,j} g - 1$. This uniqueness concerns a fixed externally supplied skeleton, not a fixed universal MRDP polynomial.

The same linearization gives several additional conclusions. A collision at causal depth h has coordinate denominators dividing D^h after integer speed normalization, where D is the least common multiple of the nonzero speed differences. Every nonempty chamber has a small positive integer point. A chamber's synchronization equalities determine its exact codimension and its perturbation behavior. For a fixed event budget, first-acceptance seeds form a finite disjoint union of such chambers, and counting them in growing integer boxes gives exact quasi-polynomials.

These conclusions solve specific representation, realizability, precision, and counting questions formulated here. They are not presented as the settlement of a named longstanding conjecture. The literature review establishes the underlying signal-machine model and related accumulation results; it does not establish historical priority for every combination of lemmas in this article. The proofs are intended to stand on their own and to be suitable for further independent review.

3 Rational signal machines and chronological skeletons

3.1 Particles, velocities, and collision rules

Definition 3.1 (Machine). A rational signal machine consists of a finite label set M , a velocity map $v : M \rightarrow \mathbb{Q}$, and a deterministic rule for every admissible incoming collision set under consideration. An incoming set contains at least two labels with pairwise distinct velocities. Its output is a possibly empty set of labels with pairwise distinct velocities. A signal of label a moves along a straight line of slope $v(a)$ until it participates in a collision.

We use a transparent default rule when an explicit rule is absent: the incoming labels reappear as outgoing labels, ordered by increasing velocity. Such a crossing still counts as a collision. Alternatively,

one may specify all relevant rules explicitly and reject a skeleton requiring an undefined rule. The attached implementation supports both conventions.

At a collision all incoming signals are consumed and the prescribed outputs are born at that space-time point. Newly born outputs are ordered by increasing velocity in the right-hand time limit. This makes them separate rather than immediately collide again. At each positive time, all spatially distinct collisions occurring at that time are processed together.

This is the fixed-speed straight-line model used in abstract geometrical computation [3]. Our arithmetic interface restricts initial configurations to finitely many signals at distinct positions and considers ordinary finite prefixes. An infinite sequence accumulating at a finite time has no automatically assigned continuation in this article.

3.2 Initial coordinates and normalization

Fix $n \geq 2$ initial labels $a_0, \dots, a_{n-1}$, in increasing spatial order, and write $d = n - 1$. Translation invariance lets us put

$$x_0 = 0, \quad x_i = g_1 + \dots + g_i \quad (1 \leq i \leq d), \quad g_i > 0. \quad (3.1)$$

Repeated labels at distinct positions are permitted. We call g the *seed geometry*; the label word is separate discrete input.

The principal chamber theorem allows real g . The Diophantine theorem uses integer g . A rational gap vector can be multiplied by a positive common denominator to become integral. Scaling all positions by $c > 0$ scales all event times by c without changing any collision rule or chronological skeleton. Thus integral seeds suffice for existence of a rational realization, but they need not preserve a separately specified physical length or time bound.

When all velocities are rational, choose an integer $L > 0$ such that $Lv(a)$ is integral for every label. In the time coordinate $\tau = t/L$, all speeds are integers $Lv(a)$. Denominator bounds below are stated in this normalized time coordinate. An arbitrary common translation in velocity is not needed.

3.3 Batches and stable signal identities

Definition 3.2 (Chronological skeleton). A finite skeleton is an ordered list of nonempty batches. Each batch is an ordered list of pairwise disjoint contiguous blocks in the current live signal list, each block having at least two members. The blocks are listed from left to right. Each block is a collision site, with its rule and outputs determined by the incoming labels. New signal identities are assigned by batch, then by site, then by increasing output velocity.

The initial identities are $0, \dots, n - 1$. A consumed identity cannot be used again; a noninitial identity must have been generated by an earlier batch. Within a single batch, outputs cannot serve as inputs of another site. These are discrete well-formedness conditions, not arithmetic constraints left for a witness to evade.

Let B be the number of batches and E the total number of collision sites. At site e , write p_e and q_e for the numbers of incoming and outgoing signals. Let m be the number of distinct velocities. Then

$$2 \leq p_e \leq m, \quad 0 \leq q_e \leq m, \quad B \leq E. \quad (3.2)$$

The quantities B , E , and causal depth are different. Many sites may share a batch; many batches may contain events unrelated to a given causal chain.

Definition 3.3 (Realization). A positive seed g realizes a skeleton σ when its actual execution has exactly the specified batches as its first B batches, with exactly the specified incoming blocks and outputs at every listed site. In particular, no earlier event and no simultaneous site or extra incoming signal may be omitted. Nothing is required about events after the final listed batch.

The empty skeleton is realized by every positive seed. It states only that no initial portion has yet been requested; it does not state that the machine never collides. A malformed skeleton is assigned the empty realization set. One can encode a statically rejected skeleton by the constant polynomial 1 if a total polynomial-producing interface is desired.

4 Eliminating every event coordinate

4.1 Intercept propagation

For each signal segment s , let v_s be its fixed velocity and write its supporting line as

$$x_s(t) = b_s + v_s t. \quad (4.1)$$

Its intercept b_s need not be its birth position: the birth may occur at a positive time. Initially, the b_s are the partial sums in (3.1), hence homogeneous integer linear forms in g .

Suppose an event contains two inputs s, r with $v_s \neq v_r$. Their formal meeting time and position are

$$t_e = \frac{b_r - b_s}{v_s - v_r}, \quad x_e = b_s + v_s t_e. \quad (4.2)$$

For an outgoing label of velocity w , its intercept is

$$b_{\text{out}} = x_e - w t_e. \quad (4.3)$$

The denominator $v_s - v_r$ is fixed by the machine, not a variable. Consequently these formulas preserve rational linearity.

Lemma 4.1 (Coordinate elimination). *For every well-formed finite skeleton, all formally reconstructed intercepts, event times, and event positions are homogeneous rational linear forms in the initial gap vector. An event with more than two incoming signals requires only additional homogeneous rational linear equalities.*

Proof. Induct on the ordered batches. Initial intercepts are linear. Equations (4.2) and (4.3) use only addition and multiplication or division by fixed rational constants. An additional incoming signal u requires $b_u + v_u t_e = x_e$, again a homogeneous linear equality. Every new intercept is then available for subsequent batches. No output of the current batch is used by another event in that batch, so this induction is acyclic. $\square$

All event coordinates are therefore *computed data*. They need not appear as free Diophantine witnesses. A certificate containing unconstrained coordinate variables would obscure both the low degree and the uniqueness that result from this elimination.

4.2 Local incidence does not imply a real computation

The incidence equations alone are insufficient. Three signals starting at $0, u, u + v$ with respective speeds $2, 0, -1$ formally give the first two a meeting time $u/2$. If $u = 3$ and $v = 1$, however, the second and third signals collide already at time 1, before $u/2 = 3/2$. An equation witnessing the later formal intersection cannot establish that the two original segments are still alive there.

There is also a boundary failure. When $u = 2v$, all three signals meet at once. A purported binary event containing only the first two signals has perfectly consistent incidence coordinates but the wrong incoming set. A compiler must distinguish strict separation from weak separation at precisely these boundaries.

For events in the same batch, choose the leftmost site's formal time as t_k . Equate all other site times in that batch to t_k , and require

$$t_k - t_{k-1} > 0, \quad t_0 = 0. \quad (4.4)$$

These conditions are necessary but still do not exclude an unlisted earlier collision. The missing ingredient is a complete spatial-order certificate.

5 A sparse certificate for complete collision chronology

5.1 Maximal adjacency intervals

Between collisions the live signals have a strict spatial order. The skeleton tells us how that order is updated: replace each incoming block by its output block, in increasing velocity order. This replacement changes only a bounded number of adjacent pairs.

Definition 5.1 (Adjacency interval). A maximal adjacency interval records an ordered pair (s, r) , left signal first, that remains adjacent in the live list from one batch boundary to a later batch boundary. The interval begins when the pair first becomes adjacent and ends when it ceases to be adjacent or when the requested prefix ends. An unchanged pair is not closed and reopened merely because an unrelated event occurs elsewhere.

Intervals created at the final batch have zero duration and need not be checked: their output order is already determined by the distinct velocities. Every recorded positive-duration interval has an affine gap

$$\Delta_{sr}(t) = b_r - b_s + (v_r - v_s)t. \quad (5.1)$$

At its initial boundary this gap is prescribed to be zero exactly when both segments are born at the same collision there. Otherwise it is prescribed to be strictly positive. At its final boundary it is prescribed to be zero exactly when both segments are consumed at the same collision there. Otherwise it is strictly positive.

Both endpoints cannot be prescribed zero for a positive-duration interval. Indeed, two distinct straight-line segments cannot separate after a common birth and then meet again without one undergoing an intervening collision. Such a proposal is rejected statically. Each surviving interval contributes exactly two linear conditions before redundant zero rows are removed.

Lemma 5.2 (Endpoint lemma). *Let $a < b$ and let f be affine on $[a, b]$. If $f(a), f(b) \geq 0$ and at least one of them is positive, then $f(t) > 0$ for every $a < t < b$.*

Proof. For $t = (1 - \lambda)a + \lambda b$ with $0 < \lambda < 1$, $f(t) = (1 - \lambda)f(a) + \lambda f(b)$ is a positive convex combination of nonnegative numbers, at least one of which is positive. $\square$

[write] Part II proves the same statement, with terminal rays added, as Lemma 21.1, and the boundary half of the soundness argument below as Lemma 21.2.

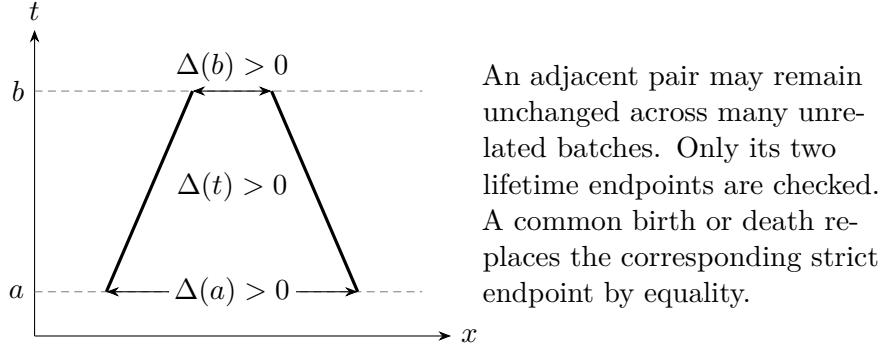


Figure 1: The affine gap between adjacent straight segments cannot become zero in the interior if its permitted endpoint conditions hold.

5.2 Soundness and completeness

Theorem 5.3 (Exact chronological chamber). *Fix a rational signal machine, an initial label word, and a finite skeleton σ . There is an effective construction of rational homogeneous linear equalities and strict inequalities whose solution set is exactly the set of real seed geometries realizing σ . All simultaneous collisions and all extra incoming signals are accounted for. Every nonempty solution set is a relatively open rational polyhedral cone in the kernel of its equality matrix.*

Proof. First reject malformed identities, overlapping or noncontiguous input blocks, undefined rules under the nontransparent convention, and incoming blocks whose velocities fail to decrease strictly from left to right. The latter condition is necessary because immediately before a multiple meeting, two incoming lines s, r ordered left to right have gap $(v_s - v_r)(t_e - t) > 0$ and hence $v_s > v_r$.

Propagate every intercept and event coordinate using (4.2)–(4.3). Add the additional incoming-incidence equalities, equal-time conditions within each batch, (4.4), initial gap positivity, and all adjacency-interval endpoint conditions described above. Every resulting expression is a homogeneous rational linear form in g .

For soundness, suppose all these conditions hold. On each positive-duration adjacency interval, the endpoint lemma gives strict separation in its interior. Hence no adjacent pair meets at an unlisted intermediate time. A meeting of nonadjacent signals would force an adjacent meeting: at the first failure of strict spatial order, at least one consecutive gap is zero. Thus no collision of any arity can occur between the listed batches.

At a batch boundary, the incidence equalities put the signals of each specified block at its reconstructed common site. Every adjacent gap not belonging to that block is strictly positive there. Consequently no additional incoming signal can join the site. The same strict conditions separate the sites of distinct blocks. They also exclude an omitted disjoint collision elsewhere in the configuration: an omitted site would have a zero gap at a boundary where strict positivity is

imposed, or would lie in the interior of an unchanged adjacency interval, where positivity has just been proved.

The deterministic rule is therefore applicable to exactly the listed incoming set at each site. Its outgoing order is increasing in velocity, so outputs separate in the right-hand time limit. Static identity validation ensures that every segment has a legitimate birth and is consumed at most once. Induction on batches now proves that the reconstructed execution is exactly the actual initial prefix.

For completeness, consider an actual execution realizing σ . The coordinate propagation agrees with its genuine intersection coordinates. The extra incidences and batch-time equalities hold, batch times strictly increase, and each endpoint gap is zero precisely at a common birth or common death of the pair. Every other endpoint gap is positive. A pair of straight segments with a common birth and a later common death cannot occur. Thus the execution satisfies every generated condition.

Finally, a finite intersection of homogeneous equality subspaces and open linear half-spaces is a relatively open convex cone in that subspace. If nonempty, its strict inequalities leave a neighborhood in the equality subspace around each point. All coefficients are rational. $\square$

5.3 The event-sparse bound

Theorem 5.4 (Linear number of chronology conditions). *Let J be the number of positive-duration maximal adjacency intervals used by the construction. Then*

$$J \leq (n - 1) + \sum_{e=1}^E (q_e + 1) \leq (n - 1) + (m + 1)E, \quad (5.2)$$

$$R_\sigma \leq 3(n - 1) + (3m + 1)E. \quad (5.3)$$

These bounds apply before deduplication or removal of identically zero equalities.

Proof. Initially there are $n - 1$ adjacent pairs. Replacing a collision block by q_e outputs creates at most $q_e - 1$ internal output pairs and two external pairs, for a total at most $q_e + 1$ when $q_e \geq 1$. If $q_e = 0$, it creates at most one bridge pair between the surviving neighbors. At a boundary or when neighboring blocks change simultaneously, fewer pairs may be created. Existing unchanged pairs need no new interval. This proves (5.2).

There are $d = n - 1$ initial gap inequalities, B positive batch-time increments, $\sum_e (p_e - 2)$ additional incoming-incidence equalities, $E - B$ equalities synchronizing the sites within batches, and $2J$ endpoint conditions. Therefore

$$\begin{aligned} R_\sigma &\leq d + B + \sum_e (p_e - 2) + (E - B) + 2J \\ &= d + \sum_e p_e - E + 2J \\ &\leq d + (m - 1)E + 2d + 2(m + 1)E \\ &= 3d + (3m + 1)E. \end{aligned}$$

$\square$

[write] The first inequality of (5.2) is Part II's Lemma 20.1; Part II's own size ledger (18.3) counts witnesses and residuals in its event-coordinate format.

The word *sparse* here refers to the number of geometric conditions, not to the number of nonzero coefficients in each eliminated linear form. A deeply propagated intercept may depend on every

initial gap. Storing all rows densely can take $O(dR_\sigma)$ coefficient slots; compiling them may have additional bookkeeping cost. Neither (5.3) nor degree two is a constant arithmetic-operation bound for an unbounded computation.

An implementation may scan the live list at each batch for simplicity. That does not change the mathematical row bound. A linked-list implementation can update adjacencies locally, while the attached small implementation prioritizes clarity and exact arithmetic over this engineering optimization.

6 Quadratic Diophantine certificates with unique slacks

6.1 Positive denominator clearing

Every nonzero rational row can be multiplied by a positive common denominator and then divided by the positive greatest common divisor of its integer coefficients. For equality rows the sign may subsequently be normalized; for strict inequality rows it may not. Negating a strict row would reverse its meaning.

Delete identically zero equality rows. An identically zero strict row makes the chamber empty. Remove duplicate rows of the same kind. Denote the resulting integer matrices by $\mathbf{E}$ and $\mathbf{G}$, with r equality rows and s strict rows. Their matrix presentation is a complete, explicit description of the polynomial, even when its expanded quadratic has many monomials.

Theorem 6.1 (Unique-slack quadratic representation). *For a nonrejected skeleton, define*

$$P(g, z) = \|\mathbf{E}g\|_2^2 + \|\mathbf{G}g - \mathbf{1} - z\|_2^2, \quad (g, z) \in \mathbb{N}_0^{d+s}. \quad (6.1)$$

Then P is an integer polynomial of total degree at most two, globally nonnegative and convex on $\mathbb{R}^{d+s}$. Moreover,

$$\exists z \in \mathbb{N}_0^s \ P(g, z) = 0 \iff g \in \mathbb{Z}^d \cap \mathcal{C}_\sigma. \quad (6.2)$$

For every accepted g , the auxiliary tuple is unique:

$$z = \mathbf{G}g - \mathbf{1}. \quad (6.3)$$

Proof. Each summand is the square of an affine linear form with integer coefficients. Thus P is a quadratic sum of squares and has Hessian $2A^\top A \succeq 0$, where A is the combined coefficient matrix of these affine forms. Over the reals, and hence over the integers, a sum of squares vanishes precisely when every squared form vanishes. Its zeros therefore require $\mathbf{E}g = 0$ and $\mathbf{G}_j g = 1 + z_j \geq 1$ for every strict row. Because these expressions are integers, this is equivalent to $\mathbf{G}g > 0$. The chamber theorem proves (6.2). The same equations force (6.3), which is a nonnegative integer tuple whenever g is accepted. $\square$

[write] Theorems 5.3 and 6.1 together are the common theorem of the two Parts (Section 1.2). Part II proves it again as Theorem 18.3, with batch times and event positions kept as natural witnesses, adds halting semantics, and strengthens uniqueness to full column rank of the witness matrix (Lemma 22.1).

All event coordinates are reconstructed from g and are absent from the auxiliary tuple. This avoids arbitrary rational encodings with multiple numerator–denominator representatives. If the seed itself is given as a rational geometry in a different interface, one must separately choose and validate an encoding or a scale; the uniqueness theorem should not be silently transferred to a noncanonical rational encoding.

6.2 What uniqueness does not assert

The polynomial depends on the externally supplied skeleton. Its number of rows and slack variables is not fixed independently of the requested history. Even though each member of this family has unique auxiliary witnesses, placing an arbitrary unbounded skeleton inside one fixed Diophantine equation is a different task. An unspecified application of MRDP to a skeleton verifier does not preserve this uniqueness for free.

The present result is therefore not a proof that all computably enumerable sets admit single-fold or finite-fold Diophantine representations. Its stronger-than-usual feature is narrower and fully explicit: for each fixed collision skeleton, projection of *all* natural zeros of (6.1) onto the initial gaps is a bijection onto the actual integer realizations. There are no additional zeros corresponding to incomplete local histories.

7 Causal depth controls denominators and coefficient size

7.1 A denominator invariant

Work in the integer-speed normalization of Section 3, and set

$$V = \max(1, \max_{a \in M} |v(a)|), \quad D = \text{lcm}\{|v(a) - v(b)| : v(a) \neq v(b)\}. \quad (7.1)$$

The least common multiple of an empty set is taken as 1; in that case no collision is possible from a simple initial configuration.

Assign depth zero to initial segments. The depth of an event is one plus the maximum depth of its incoming segments. Every output segment inherits the depth of its birth event. This is causal depth, not the chronological index of the batch.

Theorem 7.1 (Causal denominator bound). *At an event of depth h , each coefficient of its time and position forms belongs to $D^{-h}\mathbb{Z}$. The same holds for every outgoing intercept. At an integer seed, the event coordinates therefore have reduced denominators dividing D^h .*

Proof. Initial intercept coefficients are integral. Suppose an event has depth h . Its incoming intercepts have denominators dividing powers of D no larger than D^{h-1} . Put them over the common denominator D^{h-1} . Formula (4.2) divides their difference by a nonzero integer velocity difference dividing D . Its time coefficients thus belong to $D^{-h}\mathbb{Z}$. The position and all output intercepts use only addition and multiplication by integer velocities, preserving that lattice. Evaluation at integer gaps preserves the same denominator divisibility. $\square$

[write] Part II proves the same divisibility for event coordinates as Lemma 19.1, with its own proof.

Only primes already present in the speed differences can appear in these coordinate denominators at an integer seed. At a rational seed, primes in its initial common denominator may also occur. Parallel collisions elsewhere do not increase the denominator exponent along an unrelated causal chain.

This is an upper bound, not an assertion that all factors of D^h occur. Cancellations and special velocity configurations can make the actual precision requirement much smaller. Section 12 exhibits an explicit case in which exponential denominator magnitude is nevertheless unavoidable.

7.2 An explicit coefficient bound

For a rational linear form, use the ℓ^1 norm of its coefficient vector. Put $K = 2 + 4V$. Initial intercepts have norm at most d .

Lemma 7.2 (Growth of propagated forms). *Every intercept of depth at most h has coefficient norm at most dK^h . Every event time and event position of depth at most h satisfies the same bound.*

Proof. Assume the input intercept norms at a depth- h event are at most $M = dK^{h-1}$. A nonzero integer speed difference has absolute value at least one, so (4.2) gives $\|t_e\|_1 \leq 2M$. Consequently

$$\|x_e\|_1 \leq (1 + 2V)M, \quad \|b_{\text{out}}\|_1 \leq (1 + 4V)M.$$

Each factor is at most K . Induction proves the claim; forms of smaller depth obey the larger bound as well. $\square$

Proposition 7.3 (Integer row height). *If the maximum event depth is h , each primitive integer chamber row can be chosen with every coefficient of absolute value at most*

$$A = 2d(1 + V)(DK)^h. \tag{7.2}$$

In particular, for a fixed machine, row coefficient bit lengths are $O(h + \log d)$.

Proof. An endpoint gap is a difference of two intercepts plus a velocity difference times a batch time. Its coefficient norm is at most $2dK^h + 2VdK^h = 2d(1 + V)K^h$. Additional incidence residuals and differences of event times satisfy the same bound. Initial gap rows also do. All these coefficients have denominators dividing D^h by Theorem 7.1. Multiplication by D^h bounds every resulting integer coefficient by (7.2). Passing to a primitive integer row can only decrease this bound. $\square$

The exponent in (7.2) bounds coefficient *magnitude*; its logarithm is linear in depth. Exact rational arithmetic can therefore remain polynomial in the externally bounded history length even when ordinary fixed-precision floating-point simulation becomes unreliable. This is not an assertion that the total number of events before acceptance has a computable bound in the initial input length.

8 Small integral realizations and finite-horizon synthesis

Theorem 8.1 (Small positive integer seed). *Let $\mathcal{C} = \{g \in \mathbb{R}^d : \mathbf{E}g = 0, \mathbf{G}g > 0\}$ be a nonempty chamber whose strict rows include $g_i > 0$. Suppose all matrix coefficients are integers of absolute value at most $A \geq 1$. Then $\mathcal{C}$ contains a positive integer point satisfying*

$$\max_i g_i \leq d^{d/2} A^d. \tag{8.1}$$

Thus every realizable collision skeleton has a realization with a seed encoding of polynomial bit length in the explicit skeleton and initial label word, for a fixed machine.

Proof. Homogeneity permits a positive scaling of any point in $\mathcal{C}$ so that every strict expression is at least 1. Hence the closed rational polyhedron

$$Q = \{g : \mathbf{E}g = 0, \mathbf{G}g \geq \mathbf{1}\}$$

is nonempty. It lies in $[1, \infty)^d$. Minimizing $\sum_i g_i$ over Q attains an optimum: intersecting with a nonempty objective sublevel gives a compact set. The optimal face is a nonempty compact polytope and has a vertex; that point is also a vertex of Q .

At such a vertex choose d independent active constraints, including sufficient equality rows. They give a nonsingular integer matrix B and a right-hand side c with entries in $\{0, 1\}$. The vertex is $B^{-1}c$. Let $\delta = |\det B|$, a positive integer. Cramer's rule shows that $g = \delta B^{-1}c$ is integral. It is positive, satisfies the homogeneous equalities, and has every strict row at least $\delta \geq 1$.

Each coordinate of g is, up to sign, a determinant in which one column of B has been replaced by c . Hadamard's inequality bounds its magnitude by $d^{d/2}A^d$; using the sharper bound on the replaced column is unnecessary. This proves (8.1). Combining it with (7.2) gives the stated bit-length conclusion. $\square$

Corollary 8.2 (Real, rational, and integer realizability coincide). *A fixed skeleton is realizable by a positive real seed if and only if it is realizable by a positive rational seed, if and only if it is realizable by a positive integer seed. Realizability is decidable by rational linear feasibility, using $Eg = 0, Gg \geq 1$.*

The equivalence is specific to homogeneous rational chambers. It should not be transferred to arbitrary nonlinear geometry. Nor does the scaled integer seed necessarily satisfy extra nonhomogeneous requirements such as an absolute prescribed distance or a fixed deadline.

Corollary 8.3 (An NP upper bound for bounded synthesis). *Fix a rational signal machine and a distinguished output label. Given an initial label word of length n and a unary bound T , the problem of whether some positive seed produces the distinguished label within a complete prefix containing at most T collision sites is in NP.*

Proof. A skeleton of at most T sites has at most T batches and at most $n + mT$ created signal identities. Since m is fixed, its discrete description has polynomial length in $n + T$. Guess such a skeleton, check its identities and rules, and check that the distinguished label appears in its prefix.

If it is realizable, Theorem 8.1 supplies a positive integer seed with polynomial bit length: $d = n - 1$, $h \leq T$, and $\log A = O(T + \log n)$ for the fixed machine. Guess that seed as well. Propagating rational forms and verifying every chamber row then takes a polynomial number of operations on polynomial-length integers. This is a polynomial-time verifier. No search over exponentially long real numbers is required. $\square$

Only an upper bound is claimed. There is no NP-hardness result here. The budget counts total sites, including all sites in the accepting batch, not elapsed physical time. A physical-time bound can contain arbitrarily many events and even an accumulation, so it is not interchangeable with T .

9 Robustness and the rank of synchronization constraints

The chamber description distinguishes two different sources of numerical difficulty. A trace can be sensitive because it demands an exact equality among initial gaps, or its exact coordinates can require large denominators even while its combinatorial pattern is robust. The latter phenomenon will occur in the accumulating clock.

Theorem 9.1 (Certified perturbation radius and codimension). *Let $\mathcal{C} = \{g : \mathbf{E}g = 0, \mathbf{G}g > 0\}$ be nonempty, with zero strict rows removed, and let $g^* \in \mathcal{C}$. Define*

$$\rho(g^*) = \min_j \frac{\mathbf{G}_j g^*}{\|\mathbf{G}_j\|_1} > 0. \quad (9.1)$$

Every perturbation $\eta \in \ker \mathbf{E}$ with $\|\eta\|_\infty < \rho(g^)$ preserves the skeleton. The chamber has relative dimension $d - \text{rank } \mathbf{E}$. It has nonempty ambient interior if and only if $\text{rank } \mathbf{E} = 0$; otherwise it has ambient Lebesgue measure zero.*

Proof. The equality constraints persist for $\eta \in \ker \mathbf{E}$. For every strict row, Hölder’s elementary norm inequality gives

$$\mathbf{G}_j(g^* + \eta) \geq \mathbf{G}_j g^* - \|\mathbf{G}_j\|_1 \|\eta\|_\infty > 0.$$

Hence the perturbed point stays in the chamber. This proves relative openness in the full equality kernel, whose dimension is $d - \text{rank } \mathbf{E}$. A proper linear subspace has empty interior and measure zero in $\mathbb{R}^d$. Conversely, when the rank is zero, the finitely many strict inequalities define an open neighborhood of each accepted point. $\square$

The radius in (9.1) is a certified lower bound, not necessarily the largest possible radius in the equality subspace. Restricting perturbations to that subspace can increase the true distance to a forbidden boundary.

One must also distinguish genuinely independent synchronization equations from identities already forced by previous dynamics. Some formal coincidences yield an identically zero row and impose no additional codimension. Therefore the statement “every multiple collision is fragile” is too strong. What matters is the rank of the *nontrivial* conditions on the input geometry. When a loader or problem specification already restricts the admissible inputs to a subspace, robustness should be measured relative to that allowed input space.

The theorem provides a useful diagnostic for a proposed geometric computer. A simulator can report not merely a diagram, but also an exact equality matrix, a synchronization rank, and a rational certified tolerance at an integer seed. Those objects separate an intended robust mechanism from an accidental coincidence created by a particular numeric sample.

10 Finite-horizon acceptance, semilinearity, and counting

10.1 Disjoint first-acceptance chambers

Fix a machine, an initial label word, a marked accepting label, and a site budget T . Acceptance means the first appearance of that label in a batch such that the total number of processed sites through that entire batch is at most T . If the initial word already contains the marked label, every seed accepts with zero events.

There are only finitely many candidate skeletons with at most T sites. Indeed, at every stage there are at most $n + mT$ identities; the rule table and possible contiguous input blocks are finite. Enumerate those skeletons that end at the first accepting batch and reject all statically invalid ones. Distinct nonempty chambers in this first-acceptance list are disjoint: for a fixed seed, deterministic complete chronology has just one first accepting prefix.

Theorem 10.1 (Finite-horizon semilinearity). *The positive integer seed geometries accepted within site budget T form an effectively constructible semilinear set. More specifically, they are a finite disjoint union of the integer points of rational homogeneous chambers of the form (2.1).*

Proof. The enumeration above is finite and effective. For each skeleton, Theorem 5.3 gives a complete linear characterization. On integer points, a strict integer inequality is equivalent to being at least 1, so each chamber is definable in Presburger arithmetic. Finite union remains Presburger-definable. The effective equivalence with semilinear sets is the classical theorem of Ginsburg and Spanier [7]. $\square$

This use of the classical equivalence is not a new proof of that theorem. The new substrate-specific input to it is the exact finite union of collision chambers, including chronology and simultaneous-event boundaries.

[write] Part II proves the same statement for its fixed schemas and event bounds, prefix and halting, as Proposition 26.1.

10.2 A single quartic for an unknown bounded first-hit skeleton

The skeleton need not remain externally supplied when the event budget is fixed. Enumerate the nonrejected first-acceptance skeletons $\sigma_1, \dots, \sigma_K$ and let their matrices be $\mathbf{E}_k, \mathbf{G}_k$. Empty chambers may remain in this list without harm. Introduce natural selectors $b_1, \dots, b_K$ and a separate slack vector z_k for each chamber. Define

$$\begin{aligned} Q_T(g, b, z) = & \left(\sum_{k=1}^K b_k - 1 \right)^2 + \sum_{k=1}^K \sum_i (b_k \mathbf{E}_{k,i} g)^2 \\ & + \sum_{k=1}^K \sum_j (b_k (\mathbf{G}_{k,j} g - 1) - z_{k,j})^2. \end{aligned} \quad (10.1)$$

For an empty family, use $Q_T = 1$.

Theorem 10.2 (Canonical first-hit quartic). *Equation (10.1) is a globally nonnegative integer polynomial of degree at most four. Its natural zeros project bijectively onto the integer seeds accepted within the fixed event budget. No Boolean equations for the selectors and no additional inactive-slack constraints are needed.*

Proof. At a natural zero, $\sum_k b_k = 1$ forces exactly one selector to be 1 and all others to be 0. For the selected chamber, the remaining equations are precisely $\mathbf{E}_k g = 0$ and $\mathbf{G}_k g - \mathbf{1} = z_k \geq 0$. For an unselected chamber, each slack residual becomes $-z_{k,j}$, forcing all of its slacks to zero. Thus the polynomial accepts exactly the union of the chamber integer points.

Those chambers are disjoint by canonical first-acceptance chronology. Therefore an accepted seed determines exactly one selected skeleton, its active slacks are uniquely determined, and every inactive slack is zero. Every residual has degree at most two before squaring, proving the quartic bound. The construction is a sum of squares and hence globally nonnegative, but it is not in general convex. $\square$

The same construction works for any finite disjoint family of chambers. If chambers overlap, it still represents their union, but a seed in several chambers has several selector witnesses. First-hit canonicalization is therefore mathematically important, not cosmetic.

Remark 10.3 ([write] Relation to a single-fold quadratic of another report). The accepted set of Theorem 10.2 is a finite union of chamber integer points, hence semilinear (Theorem 10.1). The collection's report *canonical-diophantine-certificates* [11] proves in its Part XI (Theorem `cdc:of:thm:singlefolds1`) that every semilinear set has a representation by an integer polynomial of degree at most two that is nonnegative on the real nonnegative orthant, with exactly one natural witness for each member, effectively from a Presburger formula. So, if only orthant nonnegativity is required, the first-hit set has a single-fold quadratic certificate. Theorem 10.2 gives something different: an explicit construction from the chamber matrices, globally nonnegative on all real inputs, whose selector and slacks have the chamber meaning. Within the globally nonnegative class a quadratic has a rational affine zero set (Theorem 13.1), and the selector products of (10.1) leave that class, as the paragraph after Corollary 13.2 says. Whether a globally nonnegative quadratic suffices for a given first-hit family is not decided here.

This theorem removes external skeleton choice for a *bounded* problem; it does not remove the bound. The number of candidate skeletons, and hence the number of selectors and slacks, may be very large and grows with T . The attached implementation provides the matrix-based quartic constructor, and checks a two-chamber first-hit example. It does not include an optimized exhaustive enumerator for all skeletons of an arbitrary event budget.

10.3 Exact quasi-polynomials rather than merely leading asymptotics

Let $A_T(H)$ count the accepted seeds in $\{1, \dots, H\}^d$, with $H \geq 1$. For a single chamber put

$$\Pi_\sigma = \mathcal{C}_\sigma \cap (0, 1]^d.$$

Its closure is a bounded rational polytope in its equality subspace. The original set is obtained from that closed polytope by excluding finitely many rational faces where one or more required strict expressions vanish. The upper box facets remain included.

Theorem 10.4 (Exact finite-horizon counting). *For fixed machine, initial word, accepting label, and event budget T , $A_T(H)$ is an exact quasi-polynomial in H for all positive integers H . It has degree at most d . Its ordinary generating function is rational and has poles only at roots of unity. Furthermore,*

$$\lim_{H \rightarrow \infty} \frac{A_T(H)}{H^d} = \sum_{\sigma} \text{vol}_d(\Pi_\sigma) \in \mathbb{Q}, \quad (10.2)$$

where the sum ranges over the first-acceptance chambers, and lower-dimensional terms have zero d -dimensional volume.

Proof. Homogeneity gives $\mathcal{C}_\sigma \cap (0, H]^d = H\Pi_\sigma$. For a nonempty chamber, the closure of Π_σ is obtained by replacing its strict homogeneous inequalities by weak ones. To see equality with this weak closure, take a small strictly feasible point with every coordinate less than 1 and form convex combinations with any weakly feasible point. Those combinations converge to the latter while preserving all strict conditions and the box.

A forbidden equality $G_j g = 0$ cuts out a face of the weak closure, since $G_j g \geq 0$ throughout it. Inclusion–exclusion over these faces expresses the lattice-point count in $H\Pi_\sigma$ as a finite integer combination of counts in dilates of closed rational polytopes. Ehrhart's rational-polytope theorem makes each term a quasi-polynomial; see Beck and Robins [8]. This also covers lower-dimensional faces. Disjointness of first-acceptance chambers permits summing these exact counts.

A quasi-polynomial of degree at most d and period p has a rational generating function whose denominator divides a power of $1 - z^p$. This proves the generating-function assertion. The normalized leading coefficient of a full-dimensional rational polytope count is its volume; removing proper faces does not change it. Equivalently, the same limit follows from Riemann sums for a bounded polyhedral set. Rational polytopes have rational volume by a rational triangulation and determinant formulas. Summing over the disjoint chambers proves (10.2). $\square$

Thus a finite-horizon seed-counting problem has a particularly rigid asymptotic form: a polynomial in H with periodic coefficient functions, not an arbitrary irregular error term. Its leading density is rational. If the initial gaps are instead sampled independently and uniformly from $(0, 1]$, the probability of finite-horizon acceptance is exactly the same rational volume sum.

The theorem does not promise an efficient procedure for enumerating all chambers or computing all their volumes. Their number can grow rapidly with T . It also does not justify interchanging an unbounded-time limit with the box-size limit. Removing the event bound can leave an infinite union of chambers, outside the finite quasi-polynomial argument.

Because each integer seed in one chamber has one slack tuple, its chamber count is also the number of complete natural zeros of its quadratic certificate with $1 \leq g_i \leq H$, even when no separate bound is placed on z . Across different skeletons this statement uses a tagged disjoint family of certificates, not a single unqualified polynomial.

11 An exact first-collision tournament

11.1 Three chambers and a complete polynomial

Consider labels A, B, C with respective speeds $2, 0, -1$, initially at $0, u, u + v$, where $u, v > 0$. Introduce two stationary output labels H, R and rules

$$\{A, B\} \longrightarrow \{H\}, \quad \{B, C\} \longrightarrow \{R\}, \quad \{A, B, C\} \longrightarrow \{R\}.$$

Use transparent rules otherwise. The question here is whether H is produced at the *first* collision batch.

The possible first events are exactly

event	condition	time
AB	$u < 2v$	$u/2$
BC	$u > 2v$	v
ABC	$u = 2v$	$v = u/2$.

(11.1)

For instance, the formal AC meeting time is $(u + v)/3$. If $u < 2v$, it is later than $u/2$; if $u > 2v$, it is later than v . It cannot be a different first event. The equality case is the triple collision.

The complete accepting chamber has the particularly small certificate

$$P_{AB}(u, v, z_1, z_2, z_3) = (u - 1 - z_1)^2 + (v - 1 - z_2)^2 + (2v - u - 1 - z_3)^2. \quad (11.2)$$

All five variables range over $\mathbb{N}_0$. At $u = v = 1$, for example, the unique auxiliary tuple is $(0, 0, 0)$. At $(u, v) = (3, 1)$, the third strict condition fails, correctly rejecting the incidence-only counterfeit. At $(2, 1)$ it also fails, correctly rejecting an omitted simultaneous third input.

For the triple-collision skeleton the compiler instead produces

$$P_{ABC}(u, v, z_1, z_2) = (u - 2v)^2 + (u - 1 - z_1)^2 + (v - 1 - z_2)^2. \quad (11.3)$$

Its equality rank is one. The two binary-event chambers have rank zero and are open in the positive quadrant; the triple-collision chamber is one-dimensional and requires exact synchronization.

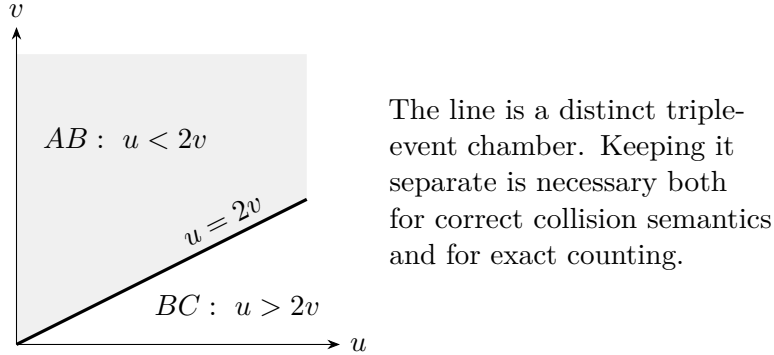


Figure 2: The first-event geometry is partitioned into two open cones and a synchronization ray. The coordinate axes are excluded.

11.2 Counting, generating function, and inversion

Let $C(H)$ count the accepting gap pairs with $1 \leq u, v \leq H$, and set $C(0) = 0$. Direct summation gives

$$\begin{aligned} C(H) &= \sum_{v=1}^H \min(H, 2v - 1) \\ &= \begin{cases} 3m^2, & H = 2m, \\ 3m^2 + 3m + 1, & H = 2m + 1, \end{cases} \\ &= \left\lfloor \frac{3H^2 + 1}{4} \right\rfloor = \frac{3H^2}{4} + \frac{1 - (-1)^H}{8}. \end{aligned} \quad (11.4)$$

For $H = 2m$, split the sum after $v = m$; the odd-number sum contributes m^2 and the remaining m terms contribute $2m^2$. The odd case follows by the same split after $m + 1$.

The first values are

$$0, 1, 3, 7, 12, 19, 27, 37, 48, 61, 75, 91, 108, 127, 147, 169, \dots$$

No assertion about an OEIS identification or novelty of this elementary sequence is needed. It is included as an exactly solvable counting test for the compiler and as an illustration of the general quasi-polynomial theorem. Its ordinary generating function is

$$\sum_{H \geq 0} C(H) z^H = \frac{z(1 + z + z^2)}{(1 - z)^3(1 + z)}. \quad (11.5)$$

This follows from the standard geometric-series derivatives applied to (11.4), or by expanding the displayed rational function.

There is also an exact inverse threshold. For every positive integer y ,

$$\min\{H \geq 0 : C(H) \geq y\} = \left\lceil \sqrt{\frac{4y-1}{3}} \right\rceil. \quad (11.6)$$

Indeed, because y is integral, $\lfloor (3H^2 + 1)/4 \rfloor \geq y$ is equivalent to $3H^2 + 1 \geq 4y$. The smooth expression inside the ceiling has the expansion

$$\sqrt{\frac{4y-1}{3}} = \frac{2}{\sqrt{3}}y^{1/2} - \frac{1}{4\sqrt{3}}y^{-1/2} - \frac{1}{64\sqrt{3}}y^{-3/2} - \frac{1}{512\sqrt{3}}y^{-5/2} + O(y^{-7/2}). \quad (11.7)$$

The binomial series supplies all further orders. The ceiling in (11.6) is essential: omitting it would incorrectly replace an integer inverse's bounded rounding oscillation by a decaying error.

11.3 Independent simultaneous sites

Take the transparent two-speed machine with $v(P) = 1$, $v(N) = -1$, and initial word P, N, P, N . With gaps $(2, 5, 2)$, the two disjoint pairs $(0, 1)$ and $(2, 3)$ collide simultaneously at time 1. A correct first batch must contain both sites. A proposed batch containing only $(0, 1)$ is rejected by the unchanged adjacency interval of $(2, 3)$, whose gap would vanish at the final boundary where positivity is required.

For general positive gaps (g_1, g_2, g_3) , this simultaneous first-batch skeleton has the single nontrivial equality $g_1 = g_3$. Changing the last gap from 2 to 3 destroys that equality. This example tests a different failure mode from the omitted third input of a multiple collision: the missing event is spatially separate, not an extra signal at the same site.

[write] Part II's hidden collision at the final cut (Section 24.2) is another instance of this failure mode, rejected by the same strict endpoint of a surviving pair (Section 1.2).

12 A four-speed clock with exponentially growing denominators

12.1 Rules and a closed-form execution

Consider the four labels

$$v(L) = 0, \quad v(P) = 2, \quad v(R) = -1, \quad v(N) = -2$$

with rules

$$\{P, R\} \longrightarrow \{N, R\}, \quad \{L, N\} \longrightarrow \{L, P\}, \quad (12.1)$$

and transparent defaults. Initially put L at 0, P at u , and R at $u + v$, with $u, v > 0$. The left boundary is stationary, the right boundary moves left at speed 1, and a particle alternately travels right at speed 2 and left at speed 2, reflecting at the boundaries via the two rules.

Write $S = u + v$ and $W = 3u + 2v$. The collision times are

$$t_{2k+1} = S - \frac{W}{3^{k+1}}, \quad k \geq 0, \quad (12.2)$$

$$t_{2k} = S - \frac{W}{2 \cdot 3^k}, \quad k \geq 1. \quad (12.3)$$

Odd collisions occur on the right boundary at position $S - t_{2k+1}$; even collisions occur at position zero.

Proposition 12.1 (Clock correctness and common chamber). *For every $u, v > 0$, equations (12.2)–(12.3) describe the actual alternating execution. The collision times strictly increase to S . Every finite alternating prefix has the whole positive (u, v) quadrant as its seed chamber.*

Proof. The first rightward meeting occurs at $t_1 = v/3$, which equals $S - W/3$. At a right-boundary collision at time t , the remaining boundary distance is $S - t$. The new left-moving particle takes $(S - t)/2$ to reach zero, leaving time-to-boundary-meeting $(S - t)/2$. From there the right-moving particle closes the remaining gap at relative speed 3, leaving two thirds of that remaining time when it next hits the right boundary. Hence a full right-to-right bounce multiplies $S - t$ by $1/3$, with the intermediate left collision having half the previous remaining time. This proves the formulas inductively.

The first time is positive. Every subsequent difference is positive, and all times are less than S . Between collisions the moving particle lies strictly between the two boundaries. The boundaries themselves meet only at S , not at any finite indexed event. There is therefore no unlisted collision in any finite prefix. The labels alternate exactly as specified by (12.1), independently of the positive gaps. This proves both correctness and the common-chamber statement. $\square$

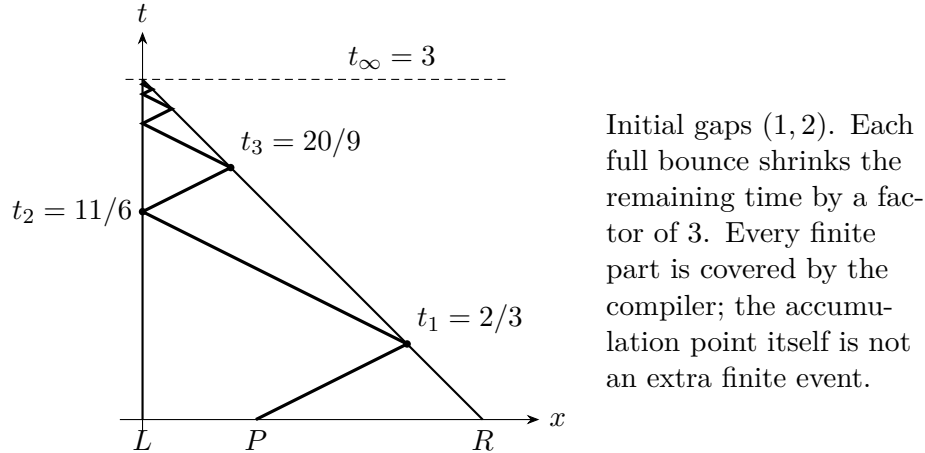


Figure 3: A four-speed accumulating clock. The drawing displays only a finite number of segments; the formulas specify all finite prefixes.

12.2 A denominator lower bound without fragile synchronization

For the particular integer seed $(u, v) = (1, 2)$, $S = 3$ and $W = 7$, giving

$$\frac{2}{3}, \frac{11}{6}, \frac{20}{9}, \frac{47}{18}, \frac{74}{27}, \frac{155}{54}, \frac{236}{81}, \frac{479}{162}, \dots$$

At event $2k$, the reduced denominator is exactly $2 \cdot 3^k$:

$$t_{2k} = \frac{6 \cdot 3^k - 7}{2 \cdot 3^k},$$

and the numerator is odd and is not divisible by 3. The causal depth of event j is j , since the reflected particle links every event to its predecessor. Thus denominator magnitude is exponential in causal depth, while its bit length is linear in depth.

Here the integer speed differences have least common multiple $D = 12$, so Theorem 7.1 correctly gives a divisible upper bound 12^j . The example does not make that upper bound sharp in its base, but it rules out a depth-independent denominator bound or a polynomial-magnitude bound for general rational signal machines.

At the same time, every finite prefix has equality rank zero and remains valid throughout the positive quadrant. Growing exact precision is therefore not evidence of a fragile collision pattern. Robust symbolic chronology and large rational denominators are compatible.

Accumulations in signal machines are established phenomena. Becker et al. [6] distinguish the behavior of small speed sets and show, in particular, that four speeds permit accumulation whereas the three-speed case with rational speed and initial-distance ratios does not. The clock is an explicit test case and a denominator-growth illustration, not a claim to have discovered accumulation. Our compiler verifies every ordinary finite prefix before S ; it assigns no transition or computational power to the limiting point at S .

[write] Part II's bounded shuttle (Proposition 25.3) uses the same mechanism on a different four-speed machine, with reduced denominators exactly 3^{k-1} (Section 1.2).

13 Why a fixed universal convex quadratic cannot follow

A natural question is whether the quadratic frontend can be compressed into one fixed polynomial with a fixed number of witnesses for arbitrary unbounded computations. Within the globally nonnegative quadratic class, there is an elementary obstruction stronger than a failure of the present construction.

Theorem 13.1 (Linear zero-set rigidity). *Let $Q(w)$ be a quadratic polynomial with rational coefficients that is nonnegative for every $w \in \mathbb{R}^N$. Its real zero set is either empty or a rational affine subspace. The projection of its nonnegative integer zeros onto any subset of coordinates is an effectively semilinear, decidable set.*

Proof. Write

$$Q(w) = w^\top A w + 2b^\top w + c$$

with A rational and symmetric. Global nonnegativity implies $A \succeq 0$. It also implies $b \perp \ker A$: otherwise the restriction to an appropriate kernel direction would be a nonconstant affine function unbounded below. Thus $b \in \text{range } A$ and the consistent rational system $A w_0 = -b$ has a rational solution.

Completing the square gives

$$Q(w) = (w - w_0)^\top A (w - w_0) + \mu, \quad \mu = c + b^\top w_0 \geq 0.$$

If $\mu > 0$, there are no zeros. If $\mu = 0$, positive semidefiniteness gives $Q(w) = 0$ exactly when $A(w - w_0) = 0$, equivalently $A w = -b$. This is a rational affine subspace.

Clearing denominators gives finitely many integer linear equations. Intersecting with $\mathbb{N}_0^N$ and projecting coordinates is an existential Presburger definition. Effective semilinearity and decidability follow from Ginsburg–Spanier and Presburger arithmetic [7]. $\square$

Corollary 13.2 (Convex-quadratic compression barrier). *An undecidable halting relation cannot be represented by one fixed globally nonnegative rational quadratic with a fixed finite number of natural parameters and witnesses, even after a computable input-loading map.*

Proof. The projected zero relation would be decidable by Theorem 13.1. Its preimage under a computable input-loading map would then be decidable as well, contradicting the assumed undecidability. $\square$

This theorem applies to global nonnegativity on *all real variables*, not merely nonnegativity on a special natural-number input slice. It does not classify arbitrary quadratic equations, arbitrary nonconvex polynomials, or cubic Diophantine representations. For example, $x - y^2 = 0$ is a quadratic equation with a nonlinear zero set, but its polynomial is not globally nonnegative and is outside the theorem.

The quartic first-hit construction in (10.1) leaves the convex-quadratic class by multiplying selectors and input-dependent linear forms. Its finite-horizon relation remains decidable for other reasons. A truly unbounded compiler must additionally encode an unbounded amount of discrete chronology into finitely many integer variables. That compression, not the local collision geometry, is where the universal Diophantine problem re-enters.

Remark 13.3 ([write] Credit). Theorem 13.1 and Corollary 13.2 are a special case of results already in Part XI of the collection’s report *canonical-diophantine-certificates* [11], which source 07 does not cite. There, Lemma `cdc:of:lem:affinezero` shows that a real polynomial p of degree at most three that is nonnegative on an open convex set O and vanishes at $a \in O$ vanishes on O exactly along $a + \ker \nabla^2 p(a)$; with $O = \mathbb{R}^N$ this is the affine zero set of Theorem 13.1. Theorem `cdc:of:thm:cubicsemilinear` shows that the full natural zero set of a *rational* polynomial of degree at most three that is nonnegative only on the real nonnegative orthant is a finite union of natural solution sets of rational affine systems, and Theorem `cdc:of:thm:classification` that the coordinate projections of such zero sets are exactly the semilinear sets, $D_2^+(k) = D_3^+(k) = \text{SL}(k)$, effectively. Those results assume less (nonnegativity on the orthant only, degree up to three, no convexity or sum-of-squares hypothesis) and give the same decidable, semilinear conclusion; that report itself notes that its theorem is stronger than the observation that a globally nonnegative polynomial cannot have odd highest degree. Corollary 13.2 therefore also holds for orthant-nonnegative polynomials of degree at most three. No novelty is claimed here for either statement. Source 07’s proof by completing the square is kept as a direct second route for the globally nonnegative quadratic case, and its clause “even after a computable input-loading map” is its own: a computable preimage of a decidable set is decidable.

13.1 The classical unbounded existence result remains available

For a finite rational initial configuration, an exact event-driven simulator can find the next collision batch by computing all positive pairwise meeting times and taking the smallest. It can then apply all rules at the sites of that batch. Repeating this procedure semidecides the appearance of a marked signal in an ordinary finite prefix. An infinite accumulating run simply causes the simulator to execute indefinitely when no earlier acceptance has occurred; no post-accumulation convention is needed for semidecidability.

Consequently the pre-accumulation finite-prefix acceptance relation, with machine and rational input encoded by natural numbers, is computably enumerable. MRDP supplies a fixed-arity Diophantine representation of that encoded relation [9, 10]. This last existence step is classical. Neither MRDP nor the present frontend automatically supplies unique witnesses, an explicit small expanded universal polynomial, or the repository’s particular operation ledger.

For a fixed universal machine with an undecidable marked-halting interface, there can be no total computable function of the encoded seed that bounds the number of sites needed by every

accepting run. Such a function, followed by exact bounded simulation, would decide halting. This explains why the bounded chamber theorems and the universal existence theorem coexist without contradiction.

[write] The last argument is the standard bounded-search argument, used for other statements in Proposition `cdc:bd:prop:nobound` of [11] and elsewhere in the collection.

14 Implementation, reproducible checks, and trust boundaries

14.1 What the attached code computes

The accompanying standard-library Python implementation uses exact `Fraction` arithmetic for every geometric calculation. The essential interfaces are:

```
Machine(speeds, rules, transparent_default=True)
simulate(machine, initial_labels, gaps, max_batches=20)
compile_skeleton(machine, initial_labels, skeleton)
UnionCertificate(tuple_of_chamber_certificates)
```

A skeleton is a tuple of batches; each batch is a tuple of input-ID tuples. For example, `((0, 1), (2, 3)),` denotes one batch with two independent simultaneous sites. Fresh identities are generated canonically by increasing outgoing velocity.

The simulator uses an all-pairs minimum-time calculation. The symbolic compiler uses intercept propagation and adjacency intervals. Their different chronology mechanisms provide a useful cross-check: the regression suite compares exact numerical prefixes against the regions predicted by the symbolic compiler on many other initial seeds.

A compiled `Certificate` contains the complete integer matrices, exact rational event forms, causal depths, interval count, and raw row ledger. Its `witnesses(gaps)` method returns the unique natural slack tuple, while `evaluate(gaps, witnesses)` evaluates the full polynomial. The `UnionCertificate` implements (10.1). Its constructor does not prove that a supplied family is globally disjoint; that is a mathematical precondition for uniqueness. At a supplied seed, its witness routine rejects multiple accepting chambers rather than silently claiming a unique answer.

The machine constructor copies and freezes the speed and rule tables. This prevents a caller's later dictionary mutation from changing the meaning of an already constructed machine. Input validation rejects unknown labels, repeated-speed outputs, malformed groups, use of dead signals, noncontiguous blocks, overlapping sites, and inappropriate incoming speed orders. These checks are part of the semantic contract.

The exported JSON matrices specify full polynomials, not just sample zeros. For example, `tournament_left.json` contains all rows of (11.2), while `first_hit_quartic_union.json` supplies every matrix and the exact selector formula for a two-chamber union. Expanding these expressions into monomials is optional and provides no extra mathematical information.

14.2 Run and reproduce

From the extracted archive's root directory, run

```
python code/run_checks.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

Python 3.10 or newer and a standard LaTeX installation are sufficient. No Python third-party dependencies are needed. The test command overwrites the generated JSON data with deterministic results, including `data/checks.json`.

[write] These are the delivered names. In this report the programs are shipped as `code/07-collision-geometry-run_checks.py` and `code/07-collision-geometry-signal_certificates.py` (with the delivered `Makefile` as `code/07-collision-geometry-Makefile`), and the seven JSON files of `data/` as `data/07-collision-geometry-*.json`. The test program imports `signal_certificates` by its delivered name and rewrites all seven JSON files in place, relative to its own directory’s parent; it must therefore be run on a copy with the delivered layout (see the README). It was so run at the write, under Python 3.14.4 on Windows: it passed with 34,560 assertions, and the seven regenerated files equal the shipped ones apart from line endings. The delivered article PDF is not shipped; this report’s PDF replaces it.

The completed run accompanying this manuscript passed **34,560 exact assertions**. Its principal coverage is summarized in Table 4. These are finite regression tests, not exhaustive verification of all machines or all input geometries.

Check family	Instances
Tournament seeds $(u, v) \in \{1, \dots, 40\}^2$	1,600
Exact seed-count boxes $H = 0, \dots, 80$	81
Exact inverse thresholds $y = 1, \dots, 2000$	2,000
Four-speed clock batches at $(1, 2)$	24
Other positive clock seeds with the same 24-batch skeleton	25
Pseudorandom rule tables (fixed random seed 20261002)	6
Distinct nonempty prefix skeletons obtained from those tables	24
Independent seed–skeleton chronology comparisons	1,536
Two-chamber quartic-union seed checks	1,600

Table 4: Finite exact test coverage. Individual assertions also check witness mutations, denominators, malformed inputs, and table immutability.

The pseudorandom cases include creation, annihilation, and transparent rules. The checked skeletons are generated from exact sample executions and then tested against other seed geometries. The suite also specifically rejects an earlier omitted collision, an omitted third input, and an omitted independent simultaneous site. It tests the clock’s exact closed-form times and denominator divisibility. For the quartic union, it checks selector validity and that inactive slacks are forced to zero.

14.3 An auditable resource ledger

Table 5 separates retained conditions from raw generated rows. The generic bound is deliberately loose; redundancy elimination can be substantial. The 24-batch clock is a useful example: its entire seed chamber reduces to only three retained strict rows, even though its geometric coordinates become increasingly complicated. One of those rows is redundant with positive gaps, but the implementation does not attempt full linear redundancy minimization.

Certificate	E	B	r	s	Raw rows	Bound
Tournament AB	1	1	0	3	7	16
Tournament BC	1	1	0	3	7	16
Tournament ABC	1	1	1	2	8	16
Two simultaneous sites	2	1	1	4	11	23
Clock, 24 batches	24	24	0	3	122	318

Table 5: Exact compiler ledger: r is the retained equality count and s the retained strict-row count. The rightmost column is $3(n-1) + (3m+1)E$, not an arithmetic-operation count.

14.4 What has and has not been verified

The prose proofs establish general claims by mathematical argument. The code and data test finite instances with exact arithmetic. No Lean or other proof-assistant verification of this article is included, and the implementation is not a formally verified compiler. The universal 13-type machine’s complete published rule table and finite-input loader have not been reimplemented in the package. The code supplies a generic machine interface and the specific worked examples only.

The repository was inspected on 2 October 2026. The observed main-branch head was

`f1edb38f93aa0a362a34b531a600909f6e74608a`.

The relevant reversible-history audit was fetched as blob

`cb31d0a108b9a19fc33da6fa89a73b604e15e260`.

No repository-wide build was performed and no repository files were changed. The current README and the older audit have different historical operation ledgers; none is used as a theorem in the present proofs. The new row counts are not comparable to a fixed-universal arithmetic count without a separate input and history compression analysis.

[write] The pin `f1edb38f9` is an ancestor of this report’s placement commit. The audit blob `cb31d0a10` is still the current `Computability/HilbertTenthProblem/Papers/1980/REVERSIBLE_FINITE_HISTORY_MODELS.md` at the write; the project README [1] has gained text since the pin and still states the 75- and 87-operation bounds. The repository’s own review of this package [14] is described in Section 1.4.

15 Further research questions and a concrete development path

[write] Table 3 pairs these directions with the questions of Part II. The first two coincide with Part II’s Questions 5 and 1.

15.1 Formalizing the chronology theorem

The most direct formalization target is the endpoint lemma followed by the sweep invariant: the live list is in strict spatial order on every open interval between batches, and its equal-position blocks at each boundary are exactly the declared sites. Stable IDs make consumption and birth finite-map properties. A Lean development should keep the real semantic statement separate from rational coefficient computation, then prove that evaluation commutes with the intercept compiler.

Finally, the sum-of-squares zero lemma and natural slack uniqueness can be linked to the repository’s Diophantine infrastructure.

A practical first milestone is a small checker for exported matrices and rational event forms. A second is a proof-producing compiler rather than a trusted Python translator. Tests that compare two unverified programs should remain auxiliary evidence, not be promoted to kernel verification.

15.2 An audited universal signal-machine frontend

Instantiate a published finite-input universal construction with its exact labels, velocities, all required collision rules, transparent defaults, input encoding, and marked acceptance convention. The 13-type construction of Durand-Lose [3] is a concrete candidate. The task is not merely to copy a picture: simultaneous three-input cases, finite workspace supply, and the loader’s correspondence with the simulated machine must be audited. The eventual theorem should state an exact equivalence between an ordinary integer program input and the existence of a compiled accepting prefix.

This would turn the present generic geometry theorem into a fully explicit end-to-end substrate result. A fixed-arity universal polynomial would still require a separate outer compression step and its own complete resource ledger.

15.3 Sharper denominator invariants

The factor D^h treats all speed differences uniformly. Can one use a velocity-labelled dependency graph to compute exact prime-valuation bounds or a smaller denominator ideal for each event? Which speed sets force uniformly bounded denominators on all integral seeds, and which permit exponential growth? The clock supplies one lower-bound mechanism, while the small-speed accumulation results [6] suggest that the interaction between arithmetic precision and global dynamics is subtler than a least-common-multiple bound alone.

A particularly concrete project is to propagate prime-valuation vectors through (4.2) and detect systematic cancellation at each rule. Any claimed lower bound must use reduced denominators, not merely the unreduced fractions produced by a particular implementation.

15.4 From few rows to small arithmetic circuits

The event-sparse theorem controls row count but permits dense coefficient vectors. Can common subexpressions in intercept forms give a substantially smaller arithmetic circuit than dense matrix evaluation? An appropriate cost model must distinguish fixed machine numerals, input-dependent arithmetic, equality tests, and the conversion from a residual system to one polynomial. The original intercept dependency graph already provides a sharing structure; the challenge is to preserve its certified semantic meaning while optimizing its arithmetic schedule.

A useful result would be a proved tradeoff among witness count, maximum degree, coefficient height, and literal addition/multiplication count. No one-dimensional comparison such as “quadratic beats a degree-169 polynomial” is meaningful when arity and unbounded-history coding differ.

15.5 Avoiding exhaustive first-hit enumeration

The quartic in (10.1) is explicit but can contain an enormous number of chambers. Can a local selector circuit encode a bounded unknown skeleton with polynomially many variables while retaining

canonical first-hit uniqueness, including all simultaneous events? A promising direction is to encode changes of adjacency directly rather than a rectangular array of all particle positions. The main obstruction is that a locally plausible selection must still describe one globally ordered and complete execution.

Even a polynomial-size bounded compiler would not solve fixed-arity unbounded compression. Keeping those two improvements distinct will make future operation claims interpretable.

15.6 Robust universality and minimal synchronization rank

For a given computational simulation, determine the rank of its required input-geometry equalities. Can the same computation be realized on a full-dimensional chamber, or does its interface genuinely require exact ratios among certain gaps? The answer must be relative to the legal input space rather than to coordinates the loader never permits to vary.

One possible invariant is the minimum independent synchronization rank among equivalent signal-machine implementations with a fixed speed set. The chamber theorem makes that invariant concrete for each finite trace, but extending it uniformly over a universal family needs a precise notion of simulation and allowed perturbation.

15.7 Counting complexity and asymptotic structure

Determine when first-hit chamber counts can be computed without enumerating all skeletons. Are there useful machine classes for which a recurrence on adjacency types produces the Ehrhart generating function directly? When do quasi-polynomial periods collapse, and which restrictions imply polynomial counts? The exact inverse in the tournament example is simple; in higher degree, inverse asymptotics must retain residue-class corrections and rounding effects rather than replacing them by an unjustified smooth error.

For unbounded acceptance, investigate convergence of finite-horizon rational densities. The increasing accepted sets give monotonicity, but a rational limit is not automatic. Quantitative tail estimates would be needed for a useful approximation theorem and for interchanging input-box and time limits.

15.8 Algebraic velocities and nonlinear motion

With velocities in a fixed algebraic number field, formal event coordinates remain linear over that field. However, positivity is no longer a rational half-space condition in the same elementary form, and integer realization by scaling is not automatic. A number-field sign certificate might extend part of the method, but the rational-cone, Presburger, and Ehrhart conclusions would require new arguments or additional hypotheses.

If signals accelerate or their velocities depend on variable parameters, (4.2) ceases to divide only by a fixed constant. This identifies a precise boundary of the present linearization. Studying which broader motion laws still admit low-degree complete chronological certificates is a more informative problem than assuming every geometric model behaves like the fixed-speed case.

15.9 Accumulation limits without fictitious infinite witnesses

The clock gives a finitely described accumulation time, but a generic finite prefix certificate cannot assert all events of an infinite run. A separate notion of a certified self-similar tail could perhaps prove a limit formula or a no-collision property beyond a given recursive pattern. Such a notion must specify the dynamics and limit semantics explicitly. It must not silently treat an infinite sequence of successful finite checks as one ordinary existential Diophantine witness.

16 Conclusion

For rational fixed-speed signal machines, the continuous coordinates of a finite computation are not the difficult nonlinear part. Once a discrete chronological skeleton is fixed, they are rational linear forms in the initial gaps. The nontrivial certificate obligation is completeness of chronology: excluding unlisted earlier meetings and accounting for every signal and every site at a simultaneous event. Maximal adjacency intervals solve that obligation with a linear number of endpoint conditions.

The resulting exact chamber description yields quadratic natural-number certificates with unique slack tuples, and finite disjoint first-hit unions yield quartics with unique selector and slack tuples. It also yields useful results about denominator growth, small integral realizations, perturbation stability, and seed-counting quasi-polynomials. The executable examples test these claims against independent exact simulation and expose the boundary cases that incidence-only encodings miss.

The unresolved universal task is sharply separated: finitely encode an unbounded discrete skeleton together with the ordinary input interface. The convex-quadratic rigidity theorem explains why this cannot be achieved by merely hiding more geometry inside one fixed globally nonnegative quadratic. ([write] The rigidity theorem is a special case of Part XI of *canonical-diophantine-certificates*; Remark 13.3.) The present contribution is a complete and auditable geometric frontend, together with a clear account of which further arithmetic and formalization steps remain.

Part II

Direct Diophantine certificates (source 11)

[write] Part II prints batch-78 manuscript 11, *Direct Diophantine Certificates for Rational Signal Machines: Finite collision schemas with unique quadratic witnesses*, “research article and exact arithmetic replay”, dated 2 October 2026; its document metadata name the author as “Research report prepared with OpenAI”. Its labels carry the prefix `smc:sd:`; its sections are Sections 17–28 here and its two appendices are Appendices C and D. It proves the theorem of Part I by a second route (Section 1.2); it is printed in full, and its lemmas that coincide with lemmas of Part I carry [write] notes. Its abstract and keywords follow.

Abstract (source 11). A prescribed finite collision history of a rational signal machine admits an explicit Diophantine certificate that retains the machine’s geometry. After external normalization, we compile each finite schema into a sum of squares of integer affine residuals. Natural zeros are exactly its realizations, and every admissible input has exactly one witness tuple. The witness coefficient matrix has full column rank, so the polynomial is strictly convex in the witnesses. A sparse construction uses event coordinates, batch times, and typed endpoints of maximal adjacency lifetimes; for bounded collision arity and fanout its size is linear in the initial support and number of events. The endpoint types enforce complete collisions, simultaneous

events, and the absence of a hidden collision at the final cut. We prove denominator divisibility and an explicit witness-height bound for the four-speed interface of a known universal signal machine. The construction is a uniform compiler for externally supplied finite schemas, not a fixed-arity representation of unbounded halting. Exact standard-library replays and fully worked examples accompany the proofs.

Keywords. Diophantine representation; signal machine; abstract geometrical computation; finite trace; unique witness; convex quadratic; semilinear set.

17 Purpose and relation to prior work

Turing-complete substrates can store and transform information in very different ways. Signal machines use a finite collection of particle types moving continuously on a line and interacting at collisions. A direct arithmetization of their geometry should explain both the prescribed events and the absence of all other events. Merely writing down line intersections leaves the latter obligation unresolved.

This article establishes that obligation for each fixed finite collision schema. Its contribution is an explicit integer certificate, including denominator control, canonical witnesses, sparse size bounds, and exact endpoint conditions. No general claim that signal-machine evolution is newly linearizable is intended. Durand-Lose’s linear real-register simulation already computes the next delay, updates gaps, and processes simultaneous events by finitely many linear branches [4, Section 4]. Our arithmetic constraints begin after a particular finite branch history has been fixed.

The substrate itself is already Turing universal. Durand-Lose gives a halting construction with 13 meta-signals and 21 *non-blank* collision rules; its four speeds are $-2, 0, 1, 2$ [3, Theorem 5 and Figure 4]. The construction uses a finite initial configuration and is separate from the paper’s weaker, non-halting cellular-automaton simulation with an infinite background. After a Galilean shift the speeds become $0, 2, 3, 4$. The rational normalized placements used in the halting arrangement can be scaled to integer positions, as proved generally below. We have not transcribed or executed that universal collision table. The accompanying replay uses small exact test machines.

The results are deliberately bounded in a precise sense. The machine, input length and labels, schema, and number of batches are compiler data. The polynomial’s arity grows with the schema. Nothing here packs unbounded execution into one fixed-arity quadratic, resolves the single-fold Diophantine representation problem, improves a universal-polynomial arithmetic-operation bound, or formalizes the proof in a proof assistant. A targeted literature check supports the stated relation to earlier work; it does not establish exhaustive novelty.

[write] Source 11 does not know source 07, printed as Part I, which proves the same theorem with all event coordinates eliminated and allows signed rational speeds; Section 1.2 compares the two routes. The repository has since reviewed this manuscript in its Hilbert-tenth-problem research tree (Section 1.4).

18 Model and exact theorem

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. Inputs and witnesses are natural numbers unless explicitly stated otherwise.

Definition 18.1 (Normalized signal machine). A machine $\mathcal{M}$ consists of a finite set of labels, a fixed speed $v(\ell) \in \mathbb{N}$ for each label, and a deterministic collision table. A legal incoming set has at

least two labels with pairwise distinct speeds. Its outgoing set also has pairwise distinct speeds and may be empty. An initial configuration consists of labels $\ell_0, \dots, \ell_{n-1}$ at positions

$$0 \leq p_0 < p_1 < \dots < p_{n-1}, \quad p_j \in \mathbb{N}.$$

Between collisions a signal follows a straight line of its label's speed. At a collision every incident signal is consumed and the designated outgoing set is emitted at that point.

If an incoming set has no rule, the collision is forbidden, unless the model explicitly completes the table by a transparent rule. Such a rule consumes the incident segments and emits the same labels, ordered by speed. The theorem works for either convention after the machine is fixed. The replay scripts use identity completion, so an omitted rule in their input is transparent, not forbidden.

We distinguish labels from *segment identities*. Each initial signal has its own identity, and each emitted signal receives a fresh one, even if a transparent rule reproduces the same label. A segment has one birth and at most one death. Initial signals are born at time zero. This distinction is essential for endpoint typing.

Definition 18.2 (Finite collision schema). Fix the machine and the initial label list. A schema $\mathcal{S}$ specifies K nonempty chronological batches. In batch i , a partition of the ordered pre-batch live list into consecutive blocks designates each nonsingleton block as an event and each singleton as a persisting signal. Every event uses a legal rule, and its incoming speeds strictly decrease from left to right. Its outputs are ordered by strictly increasing speed. All replacements in a batch are performed atomically. Segment births precede their deaths by at least one batch.

The schema fixes no event coordinates or durations. Its events within a batch are ordered from left to right by their incoming blocks. Zero-output events are retained as events. A schema with $K = 0$ has no events. No initial collision markers, infinite initial support, or accumulation events are included.

A *prefix realization* has exactly these first K nonempty collision batches and ends immediately after batch K . The $K = 0$ prefix is the initial configuration alone. A *halting realization* additionally has no future collision. Halting does not require all signals to disappear or stop moving. There is no arbitrary terminal observation time.

Let E be the number of events, $I = \sum_e |\text{in}(e)|$ the number of incoming incidences, r the largest incoming arity, and s the largest fanout. Set $n_- = \max(n - 1, 0)$ and

$$D = \text{lcm}\{|v - w| : v, w \text{ are machine speeds, } v \neq w\}, \quad B = D^K, \quad (18.1)$$

with the empty least common multiple equal to one.

[write] Theorem 18.3 is, for prefix semantics, the theorem of Part I's Theorems 5.3 and 6.1, in a different witness format: natural absolute positions as input, and batch times, event positions and slacks as witnesses on the grid B . Its halting version, the full column rank and the size ledger (18.3)–(18.4) are not in Part I. Source 11's proof is printed in full as the second route.

Theorem 18.3 (Sparse canonical certificate). *There is an effective compiler which, for every such fixed schema, produces integer affine residuals $r_1, \dots, r_R$ in the input coordinates $\mathbf{p}$ and witnesses $\mathbf{y} \in \mathbb{N}^W$, and a polynomial*

$$P_S(\mathbf{p}, \mathbf{y}) = \sum_{a=1}^R r_a(\mathbf{p}, \mathbf{y})^2.$$

For prefix semantics, a natural zero exists exactly when $\mathbf{p}$ realizes the schema. For halting semantics the same statement holds after the terminal conditions below are included. For every admissible input the witness is unique. The witness coefficient matrix has full column rank; consequently, if $W > 0$, the Hessian in the witness variables is positive definite.

If L denotes the number of recorded adjacency lifetimes, including terminal rays when present, then

$$L \leq n_- + \sum_e (|\text{out}(e)| + 1), \quad (18.2)$$

$$W \leq E + 2K + 2L + n_-, \quad R \leq I + K + 2L + n_-. \quad (18.3)$$

In particular,

$$W \leq (2s + 5)E + 3n_-, \quad R \leq (r + 2s + 3)E + 3n_-. \quad (18.4)$$

The polynomial has degree at most two, and degree exactly two whenever its witness vector is nonempty.

Syntactically illegal schemas are rejected before compilation. For the halting version, a final live order with a negative adjacent relative velocity is likewise rejected as impossible. One can instead output the constant polynomial one for these cases; the stated resource ledger concerns the structurally admitted construction. At $K = 0$, the prefix compiler may use only input-order checks. For $n \leq 1$ it has no witnesses and returns zero. This harmless degenerate case is not a strictly convex function on a positive-dimensional space.

19 Rational normalization and denominator control

19.1 Normalization is an external interface

Suppose the original speeds are rational numbers $u(\ell)$. Choose a rational $c \leq \min_\ell u(\ell)$ and a positive integer m such that every $m(u(\ell) - c)$ is an integer. For a fixed rational input choose a rational translation a and positive integer q so that every $q(x_j + a)$ is a nonnegative integer. The invertible coordinate change

$$X = q(x + a - ct), \quad T = \frac{q}{m}t \quad (19.1)$$

preserves simultaneous incidence, spatial order at each time, positive time direction, and the collision rules. Its speeds are

$$\frac{dX}{dT} = m(u(\ell) - c) \in \mathbb{N},$$

and its initial coordinates are natural numbers. Thus every finite rational configuration can be put into the theorem's interface.

The constants a, c, m, q are chosen *outside* the existential witness tuple. For a varying input family with fixed denominator and shift, the natural input $p_j = q(x_j + a)$ is the declared coordinate encoding. Allowing an arbitrary additional scale as a witness produces many proportional encodings; allowing raw variable numerators and denominators introduces products when denominators are cleared. Neither change preserves the present canonical, jointly quadratic claim without a new argument. Inputs at arbitrary real coordinates are outside this interface.

[write] Part I needs no Galilean shift: it allows signed integer speeds after the time rescaling $\tau = t/L$ (Section 3) and takes the gap vector as input. With $m = L$ the two normalizations give the same speed differences, hence the same D (Section 1.2).

19.2 Causal depth and a fixed grid

Assign depth zero to initial births. The depth of event e is one plus the maximum depth of the birth events of its incoming segments, with initial births contributing zero. Every birth lies in an earlier batch, so the maximum event depth H satisfies $H \leq K$.

Lemma 19.1 (Denominator divisibility). *([write] The same as Part I, Theorem 7.1, there stated for the coefficient forms in the initial gaps; source 11's proof is kept as a second route because it works with event-coordinate witnesses.) Every event time and position of depth at most h has reduced denominator dividing D^h . In particular D^H , and hence $B = D^K$, clears every event coordinate and time. It also clears the position of any live segment sampled at an event time.*

Proof. Initial birth positions and times are integers. Suppose an event of depth h has incoming lines

$$x = b_1 + v_1(t - a_1), \quad x = b_2 + v_2(t - a_2), \quad v_1 \neq v_2.$$

All birth coordinates have denominators dividing D^{h-1} by induction. The intersection time is

$$t = \frac{b_2 - v_2 a_2 - b_1 + v_1 a_1}{v_1 - v_2}. \quad (19.2)$$

The numerator's denominator divides D^{h-1} , while $|v_1 - v_2|$ divides D . Thus the reduced denominator of t divides D^h . Substitution into either line gives the same assertion for its position. A multi-signal event is pinned by any two incoming speeds; the remaining incidences are additional equalities. Finally a sampled position is an integer-linear combination of a birth position, birth time and sample time, all cleared by D^H . $\square$

This lemma bounds denominator divisibility, not numerator magnitude. Its practical advantage is that B is a computable constant of the schema. The compiler can use the sharper D^H if desired; the supplied implementation uses D^K .

20 The sparse construction

20.1 Geometry variables and flight equations

Use a natural variable T_i for the scaled time of batch i , with $T_0 = 0$, and a natural variable X_e for the scaled position of each event. Introduce $d_i \in \mathbb{N}$ and require

$$T_i - T_{i-1} - d_i - 1 = 0 \quad (1 \leq i \leq K). \quad (20.1)$$

Thus each batch occurs at a strictly later time on the B -grid.

For a segment σ of speed v_σ , let $b(\sigma)$ be its birth batch. Define its scaled birth-position expression A_σ to be Bp_j for initial segment j , and X_f if it is emitted at event f . Its position at any relevant batch time is the affine expression

$$Z_\sigma(T_i) = A_\sigma + v_\sigma(T_i - T_{b(\sigma)}). \quad (20.2)$$

If σ is consumed by event e in batch i , impose the flight residual

$$X_e - Z_\sigma(T_i) = 0. \quad (20.3)$$

There are I such residuals. All coefficients are integers because the speeds and B are fixed integers. No product of two unknowns appears.

Input order is encoded, rather than silently assumed, by natural slacks u_j and residuals

$$p_{j+1} - p_j - u_j - 1 = 0 \quad (0 \leq j < n - 1). \quad (20.4)$$

An explicitly restricted ordered-input domain permits these residuals and witnesses to be omitted. We keep them in every count.

20.2 Maximal adjacency lifetimes

Maintain the ordered live list combinatorially. A consecutive ordered pair of segment identities has a lifetime beginning at the first boundary at which it becomes adjacent and ending at the boundary at which it ceases to be adjacent. A pair that remains adjacent across several batches has one lifetime, not a new copy in every slab. Perform all replacements in a simultaneous batch before creating new pairs; pairs appearing only during an arbitrary sequential processing order are never recorded.

For a prefix, close each still-active positive-duration lifetime at the final cut T_K . A pair newly born at batch K has no positive-duration finite lifetime and is omitted. For halting semantics, still-active pairs receive infinite terminal lifetimes instead, including pairs born at the final batch. The initial $K = 0$ prefix needs no finite lifetimes.

For a pair (σ, τ) , left to right, its scaled gap is the affine function

$$G_{\sigma\tau}(T) = Z_\tau(T) - Z_\sigma(T), \quad (20.5)$$

whose slope is the fixed relative velocity $v_\tau - v_\sigma$.

Lemma 20.1 (Lifetime count). ([write] *The same as the first inequality of (5.2) in Part I, Theorem 5.4; source 11's proof is kept as a second route because it works with event-coordinate witnesses.*) *The number L of recorded lifetimes satisfies (18.2).*

Proof. There are at most n_- initial adjacent pairs. Replacing one consecutive event block by k outputs creates at most $k + 1$ new pairs: at most $k - 1$ internal output pairs and two outer-neighbor pairs when $k \geq 1$, or at most one bridge pair when $k = 0$. Boundary events can create fewer. A sequential accounting of the disjoint replacements in a batch therefore counts at most $\sum_e (|\text{out}(e)| + 1)$ creations. Some counted pairs may be transient and disappear within the same batch; atomic processing removes those and cannot increase the count. Every lifetime has exactly one creation, including any terminal lifetime. Closing a lifetime does not create an additional one. $\square$

20.3 Typed endpoint constraints

For a finite lifetime from batch a to batch $b > a$, impose two constraints:

1. At the lower endpoint, impose $G(T_a) = 0$ if and only if both segments were emitted by the *same event* in batch a . Otherwise impose $G(T_a) = h^- + 1$ with a fresh $h^- \in \mathbb{N}$.
2. At the upper endpoint, impose $G(T_b) = 0$ if and only if both segments are consumed by the *same event* in batch b . Otherwise impose $G(T_b) = h^+ + 1$ with a fresh $h^+ \in \mathbb{N}$.

The “if and only if” here specifies which residual the compiler emits from the schema; it is not a new logical connective in the polynomial. Permitted equality means an equality residual, not an arbitrary nonnegative gap. In particular, a surviving pair at the final cut receives a *strict* upper gap.

For a terminal ray starting at batch a , require its relative velocity to be nonnegative, a check on machine constants. Impose a zero initial gap only for two same-event outputs at that starting boundary; impose $G(T_a) = h + 1$ otherwise. Same-event outputs have strictly increasing speeds, so a zero-gap terminal pair automatically has strictly positive relative velocity. A terminal pair with zero relative velocity must therefore start with positive gap.

Every stated condition is either an integer affine residual or a finite compile-time structural check. Define P_S by squaring and adding the residuals (20.1), (20.3), (20.4), and all endpoint residuals.

21 Proof of geometric correctness

21.1 Why endpoint types suffice

Lemma 21.1 (Positive interior gap). *([write] The same as Part I, Lemma 5.2 (the endpoint lemma) together with the static rejection of a common birth and a common death that precedes it; source 11's proof is kept as a second route because it works with event-coordinate witnesses, and it adds terminal rays.) Every finite lifetime satisfying its endpoint residuals has strictly positive gap in its open time interval. Every admitted terminal lifetime has strictly positive gap at every time after its start.*

Proof. A finite gap is affine. Its endpoint values are nonnegative. If either is strictly positive, every proper convex combination of the endpoint values is strictly positive. If both are zero, the endpoint types say that the pair shares its birth event at the lower boundary and its death event at the upper boundary. Birth ordering gives $v_\sigma < v_\tau$, so

$$G(T_b) = G(T_a) + (v_\tau - v_\sigma)(T_b - T_a) > 0,$$

a contradiction. Thus at least one endpoint is strict, and no extra endpoint-sum residual is needed. On a terminal ray the nonnegative slope preserves a strict initial gap; a zero initial gap has strictly positive slope and becomes positive immediately. $\square$

It is important that this argument uses segment identities and the exact start and end boundaries. Merely allowing two nonnegative endpoint gaps, even with positive sum, does not exclude an unrecorded meeting at the upper endpoint.

Lemma 21.2 (Complete boundaries). *At every batch satisfying the residuals, each designated incoming block meets at its event point, distinct designated events have distinct ordered positions, and no persisting signal or signal of another event also meets that point. The post-batch order is valid; its only zero gaps are between outputs of the same event.*

Proof. Before the batch, every adjacent pair belongs to a lifetime that has existed for a positive duration. For a pair in one event block, the flight equations place both segments at that event. Any pair across two distinct pre-batch blocks is not consumed by the same event. If its lifetime ends at this boundary, the strict upper endpoint gives a positive gap. If it persists across the boundary and this is an interior point of its lifetime, Lemma 21.1 gives a positive gap. If the boundary is the final cut, a surviving pair has a strict upper endpoint, or lies on an admitted terminal ray. These cases exhaust the pre-batch neighboring pairs.

All positions in an incoming block are equal by the flight equations, whereas successive blocks are strictly separated. Hence the designated blocks are precisely the maximal equality classes at that time. In particular a third incident signal cannot be omitted, two designated events cannot

coincide, and a completely erased event still has a position in the strict block order. Replacing each event block at its own point by the prescribed speed-ordered outputs therefore preserves weak spatial order. Across distinct surviving blocks the gap stays strict; within one output fan the gap is zero and the increasing speeds separate it immediately. If erased blocks disappear, the new bridge gap spans one or more former strict block separations, so is positive. The replacements are atomic and introduce no additional intermediate configuration. $\square$

This proof also explains why final-born prefix pairs can be omitted. Their boundary order is already certified by the incoming block partition and flight equations. Immediately after that boundary, a common output fan separates and all other gaps remain positive for a sufficiently small interval. A prefix does not assert the absence of collisions later in that interval's future.

21.2 Soundness and completeness

Proof of the existence assertion in Theorem 18.3. For soundness, suppose $P_S(\mathbf{p}, \mathbf{y}) = 0$ with natural inputs and witnesses. Each square is nonnegative, so every residual is zero. The input packet gives strict initial order. Time residuals give strictly increasing batch times. Divide all scaled times and event positions by B and draw the prescribed segments using (20.2).

On each open slab, the combinatorial live list is fixed and every neighboring pair is in an active lifetime. Lemma 21.1 excludes all adjacent meetings. A meeting of nonadjacent signals would force an adjacent equality in the ordered list at or before that meeting, so there are no other meetings either. At the next batch, Lemma 21.2 proves that the designated events are exactly all collisions and contain every incident signal. Their legal rules are the fixed schema rules. Induction across the K batches gives precisely the prescribed prefix. For halting semantics every final adjacent gap remains positive for all future time by the terminal-ray argument, so no future collision occurs.

For completeness, start with an actual realization. Lemma 19.1 puts every event time, event position and endpoint sample on the B -grid. Assign their scaled values to T_i and X_e , and set $d_i = B(t_i - t_{i-1}) - 1$. All are natural. Input gaps determine natural u_j . Each lower or upper endpoint at which the schema prescribes equality is an actual common birth or death point. Every other endpoint is strictly separated by the definition of a complete collision history; its positive scaled integer gap minus one supplies the unique natural slack. An actual halting configuration has nonnegative adjacent relative velocities, since a negative one would eventually cause a meeting, and its terminal initial gaps have exactly the required types. All residuals vanish. $\square$

The argument covers simultaneous multi-signal collisions, disjoint events in the same batch, annihilation, transparent collisions and empty final configurations. Its dependence on complete blocks and typed boundaries is substantive, not a cosmetic choice of encoding.

22 Canonical witnesses and strict convexity

Uniqueness of integer witnesses alone would not prove strict convexity. We establish the stronger linear-algebra statement directly.

Lemma 22.1 (Full column rank). *Write the residual vector as*

$$\mathbf{r}(\mathbf{p}, \mathbf{y}) = A\mathbf{y} + C\mathbf{p} + \mathbf{c}.$$

For every structurally admitted schema, the matrix A has trivial kernel over $\mathbb{R}$, even if the schema has no realization for the chosen input.

Proof. Let $A\Delta\mathbf{y} = 0$ and hold the input fixed. Initial birth positions have zero variation. Induct on batches. Assume all earlier times and event positions have zero variation. Choose any event e in batch i and two of its incoming segments σ, τ with distinct speeds. The homogeneous flight equations are

$$\Delta X_e - v_\sigma \Delta T_i = 0, \quad \Delta X_e - v_\tau \Delta T_i = 0,$$

because both segments were born in earlier batches. Their difference gives $(v_\sigma - v_\tau)\Delta T_i = 0$, hence $\Delta T_i = 0$ and $\Delta X_e = 0$. Every other event in the batch then has zero position variation from any one of its flight equations. Equation (20.1) gives $\Delta d_i = 0$. Induction fixes all geometry variables and time slacks.

Each remaining witness is an input-order or endpoint slack with coefficient -1 in its own defining residual. With the geometry and input fixed, that residual forces its variation to be zero. Thus $\Delta\mathbf{y} = 0$. $\square$

Corollary 22.2 (Canonical zero and Hessian). *For fixed input there is at most one real solution of the residual system, and hence at most one natural zero of P_S . The Hessian in witness coordinates is $2A^\top A$, positive definite when $W > 0$.*

Proof. Two solutions differ by a vector in $\ker A$, which is zero by Lemma 22.1. Moreover $\mathbf{z}^\top (2A^\top A)\mathbf{z} = 2\|A\mathbf{z}\|^2 > 0$ for every nonzero $\mathbf{z}$. Existence for admissible inputs was proved above. $\square$

Strict convexity is asserted in the witness variables with the input held fixed. Joint strict convexity in input and witness coordinates is neither needed nor generally true. The compiler introduces no witness for a schedule choice, denominator choice or arbitrary final observation time. Adding any of those could destroy the canonical property.

Proof of the size assertions in Theorem 18.3. Geometry uses E event positions, K batch times and K time slacks. A finite lifetime needs at most two slacks and two residuals; a terminal lifetime needs at most one of each. Input order uses n_- slacks and residuals. Flight and time residual counts are I and K . This proves (18.3). Lemma 20.1, $I \leq rE$, $K \leq E$ and $|\text{out}(e)| \leq s$ give (18.4). Squaring affine residuals gives degree at most two. If $W > 0$, full column rank forces a nonzero quadratic part. $\square$

The ledger counts variables and affine residuals. With a fixed machine, every residual has bounded width: an endpoint gap uses only the birth positions and times of two segments, one endpoint time, and possibly a slack. A linear-size arithmetic circuit follows if integer constants are treated as available coefficients. This does not give a literal gate ledger under a model that charges separately for constructing large constants, input loading, multiplication conventions or natural-to-positive conversion. In particular, no improvement to previously studied 75-operation or 87-operation constructions is asserted here.

23 A full-layer reference construction

A denser construction is useful for cross-checking. In slab i , let N_i be the number of pre-batch signals and g_i the number of blocks, including singleton through-signals. Introduce natural scaled

end coordinates $C_{i,j}$, a duration slack d_i , and one gap slack between successive blocks. The scaled start coordinate $A_{i,j}$ is Bp_j initially or an alias of an earlier endpoint. Require

$$\begin{aligned} C_{i,j} - A_{i,j} - v_j(d_i + 1) &= 0 && \text{for every live signal,} \\ C_{i,j} - C_{i,\text{first}(b)} &= 0 && \text{inside each collision block } b, \\ C_{i,\text{first}(b')} - C_{i,\text{first}(b)} - h - 1 &= 0 && \text{for successive blocks } b, b'. \end{aligned}$$

All blocks, including zero-output collisions, participate in the last row. The exact counts, with input-order checks, are

$$W = n_- + \sum_i (N_i + g_i), \quad R = n_- + \sum_i (2N_i - 1). \quad (23.1)$$

Indeed each slab uses $N_i + 1 + (g_i - 1)$ witnesses, and $N_i + (N_i - g_i) + (g_i - 1)$ residuals.

The two boundary lists have nonnegative adjacent gaps, with strict separation across ending blocks. A pair with zero gap at its starting boundary is a same-event output pair with increasing speeds; it cannot meet again at the ending boundary after positive time. Affineness therefore excludes all interior collisions. The full ending partition supplies completeness at each boundary. Conversely the denominator lemma gives a witness from every realization. The same two-incoming-speed induction fixes each duration, then all endpoints and gap slacks, and proves full witness rank. A halting certificate additionally checks that the final live speeds are nondecreasing; the final boundary order is already certified.

This reference compiler stores coordinates for every live signal at every layer, even if that signal survives unchanged through many layers. The sparse compiler avoids these repetitions by storing only event coordinates and maximal adjacency lifetimes. Both implementations share an exact simulator and small arithmetic utilities; agreement between them is useful evidence, but is not independent formal verification of the common simulator.

24 Worked examples

24.1 One annihilation with its complete quadratic

Consider labels R, S with speeds $2, 0$ and the rule $\{R, S\} \mapsto \emptyset$. Let the initial positions be $p_0 < p_1$, and specify one event. Here $D = B = 2$. Write T, X, d for its scaled time, position and duration slack, u for input order and h for the lower adjacency slack. The compiler gives

$$\begin{aligned} P &= (p_1 - p_0 - u - 1)^2 + (T - d - 1)^2 \\ &\quad + (X - 2p_0 - 2T)^2 + (X - 2p_1)^2 \\ &\quad + (2p_1 - 2p_0 - h - 1)^2 + (2p_1 - 2p_0 - 2T)^2. \end{aligned} \quad (24.1)$$

The last row is the zero upper gap of the two segments consumed by the same event. Its redundancy with flight equations is harmless. Every $p_0 < p_1$ has the unique witness

$$T = p_1 - p_0, \quad X = 2p_1, \quad d = u = p_1 - p_0 - 1, \quad h = 2(p_1 - p_0) - 1.$$

For $(p_0, p_1) = (0, 1)$ the tuple (T, X, d, u, h) is $(1, 2, 0, 0, 1)$, corresponding to physical time $1/2$ and position 1 . There are five witnesses and six residuals. The final state is empty, so the prefix is also halting. Changing any one witness while fixing the others cannot keep all residuals zero.

24.2 A hidden collision exactly at the final cut

Use the four-speed interface $\{0, 2, 3, 4\}$, with labels a, b, c, d respectively, and the annihilation rule $\{d, a\} \mapsto \emptyset$. Start with

$$(\ell_0, \ell_1, \ell_2, \ell_3) = (d, a, d, a), \quad (p_0, p_1, p_2, p_3) = (0, 4, 10, 14).$$

There are two simultaneous collisions at physical time 1, at positions 4 and 14. Now submit a false one-batch schema containing only the first event and marking the last two signals as through-signals. Since $B = 12$, the first event's flight equations force $T_1 = 12$ and $X_0 = 48$.

The omitted pair's gap is

$$G(T) = 12(14 - 10) - 4T = 48 - 4T.$$

It is positive before the cut, has lower value 48, and reaches zero at the cut. Thus conditions of the form $G(0) \geq 0$, $G(T_1) \geq 0$, and $G(0) + G(T_1) > 0$ would accept its endpoints. Our upper endpoint is strict because the pair survives rather than dying in one designated event. Its residual becomes

$$G(T_1) - h^+ - 1 = -h^+ - 1,$$

which cannot vanish for $h^+ \in \mathbb{N}$. The false schema has no certificate. The correct simultaneous schema consumes both pairs atomically and has event positions 48 and 168 on the scaled grid.

[write] Part I's independent simultaneous sites (Section 11.3) are another instance of this failure mode: an omitted, spatially separate simultaneous collision, rejected by the strict endpoint of the surviving pair (Section 1.2).

24.3 An outgoing fan at a finite cut

Replace the annihilation rule in the preceding setup by $\{d, a\} \mapsto \{b, c\}$. Each collision emits speeds 2 and 3. At the final cut the two members of each fan have zero gap and immediately separate; they need no zero-duration finite lifetime. In a halting certificate they instead receive terminal rays with zero starting gap and positive relative velocity. The gap between the two different fans is strict at the cut. Its relative velocity is $2 - 3 = -1$, so that terminal ray is rejected: the two fans will interact later. This separates valid finite-prefix geometry from the stronger no-future-collision requirement.

25 Witness height for the four-speed interface

Let the normalized machine speeds lie in $\{0, 2, 3, 4\}$, and let M_j be the largest live position immediately after batch j , with $M_0 = \max_j p_j$ and $\max \emptyset = 0$. Let t_j be elapsed physical time, with $t_0 = 0$.

Lemma 25.1 (Geometric growth). *Whenever batch $j + 1$ exists, its delay satisfies $\delta_{j+1} \leq M_j$. All positions at that boundary, including erased event positions, are at most $5M_j$. Consequently*

$$M_j \leq 5^j M_0, \quad t_j \leq M_0 \frac{5^j - 1}{4}.$$

Proof. A next collision must be approached by an adjacent closing pair. The pair has a strictly positive gap g and positive integer closing speed $c \geq 1$. Because all live positions are nonnegative,

$g \leq M_j$; its meeting delay is $g/c \leq M_j$. The earliest collision delay is no larger. During this interval every signal advances by at most $4M_j$, so every pre-collision position is at most $5M_j$. Outputs remain at those event positions and deletion cannot increase the maximum. Iterate this bound and sum $\delta_i \leq 5^{i-1}M_0$ to obtain the stated geometric series. $\square$

The argument concerns an existing next batch. Empty or singleton configurations have none. Nor does a valid configuration whose maximum position is zero: any coincident outgoing fan has strictly increasing speeds and separates. This avoids dividing by a nonexistent positive gap.

Theorem 25.2 (Canonical height). *For $B = 12^K$, every canonical input coordinate and witness of the sparse or full-layer compiler is at most*

$$60^K \max(1, M_0). \quad (25.1)$$

Each such integer therefore has bit length $O(K + \log(M_0 + 1))$.

Proof. For the full speed set, the nonzero absolute differences are 1, 2, 3, 4, whose least common multiple is 12. The same 12 is a valid clearing modulus for a subset. Up to batch K , every live sample and every event position is at most $5^K M_0$, including positions subsequently erased. Multiplication by 12^K bounds them by $60^K M_0$. Every endpoint gap is the nonnegative difference of two positions in this range, so its slack is smaller. Input-order slacks are at most M_0 . The time bound is

$$12^K t_j \leq 12^K M_0 \frac{5^K - 1}{4} \leq 60^K M_0,$$

and the scaled duration minus one is no larger than the corresponding scaled time. The empty and all-zero cases are covered by $\max(1, M_0)$; if $K = 0$, all remaining slacks are initial-gap slacks bounded by M_0 . Taking binary logarithms proves the bit bound. $\square$

This bounds actual zero witnesses, not arbitrary trial assignments. Degree two alone would supply no such height guarantee. Likewise, the bound is exponential in batch count K as a numerical magnitude while linear in K as a bit length. For a fixed machine with maximal speed V , the same argument gives $M_j \leq (1 + V)^j M_0$; the four-speed constant above is the concrete interface relevant here.

[write] Part I bounds the chamber's row coefficients (Proposition 7.3) and the size of *some* realizing seed (Theorem 8.1); Theorem 25.2 bounds the canonical witnesses of every realizing input. The two bounds measure different things and neither implies the other.

25.1 Exponential denominator growth can be necessary

Proposition 25.3 (A bounded geometric shuttle). *There is a fixed four-label machine with integer initial positions whose $2k$ -batch prefix contains a time of reduced denominator 3^{k-1} , although every event lies in $0 < t < 2$ and $0 < x < 6$. Thus a common integer grid for exact event geometry may require exponentially large scale in the number of batches.*

Proof. Give labels L, R, F, B speeds 3, 2, 4, 0, respectively, and use rules

$$\{F, R\} \mapsto \{B, R\}, \quad \{L, B\} \mapsto \{L, F\}.$$

Start with L, F, R at positions 0, 1, 2. The two walls follow $x_L(t) = 3t$ and $x_R(t) = 2 + 2t$, so their separation is $2 - t$ until they meet at $t = 2$. The first right collision occurs at $t_{R,1} = 1/2$ and the

first left collision at $t_{L,1} = 1$. At a left collision at time $a < 2$, the outgoing speed-four shuttle catches the right wall after $(2 - a)/2$, giving right time $r = 1 + a/2$. Its speed-zero replacement is then caught by the left wall after $(2 - r)/3$, giving the next left time

$$a' = r + \frac{2 - r}{3} = \frac{4}{3} + \frac{a}{3}.$$

These positive delays place the shuttle strictly between the walls between alternating collisions; there are no other earlier meetings. The recurrence and first collision give

$$t_{L,k} = 2 - 3^{1-k}, \quad t_{R,k} = 2 - \frac{3}{2} 3^{1-k} \quad (k \geq 1).$$

The first six batch times are $1/2, 1, 3/2, 5/3, 11/6, 17/9$. At the k th left event the numerator of $(2 \cdot 3^{k-1} - 1)/3^{k-1}$ is coprime to three, so the reduced denominator is exactly 3^{k-1} . Event positions lie on a wall before time two and hence are below six. Every finite prefix is valid. $\square$

This example supplies a lower bound for common-grid geometry encodings: after $2k$ batches a clearing scale is divisible by 3^{k-1} , and its scaled last time requires $\Omega(k)$ bits. It makes the linear event-count dependence in the witness-bit bound qualitatively sharp for this type of encoding; it is not a lower bound on arbitrary Diophantine representations. The collision times approach a Zeno accumulation at $t = 2$. We assert nothing about a continuation at or beyond that accumulation, which is outside the finite-schema theorem. The replay checks 64 batches with exact fractions.

[write] Part I's four-speed clock (Section 12, Proposition 12.1) is the same shuttle mechanism on a different machine, with speeds $0, 2, -1, -2$ and reduced denominators exactly $2 \cdot 3^k$ at event $2k$ for the seed $(1, 2)$. The worked-example receipt shipped as `data/11-signal-certificates-expected-worked_examples_receipt.json` records the last left denominator of the 64-batch prefix as $617673396283947 = 3^{31}$, which is 3^{k-1} for $k = 32$.

26 Fixed-schema arithmetic and the unbounded boundary

Proposition 26.1 (Semilinear feasible inputs). *([write] The same as Part I, Theorem 10.1, for prefix semantics; source 11's proof is kept as a second route because it works with event-coordinate witnesses, and it adds the halting case.) For a fixed normalized machine, fixed initial label list and fixed schema, the set of natural input positions admitting a prefix certificate is semilinear. The same holds for the halting certificate. For a fixed total-event bound, the union over all schemas with that bound is also semilinear, provided the initial length and labels are fixed.*

Proof. A zero of a sum of squares is exactly a common zero of its affine residuals. Existence of natural witnesses is therefore an existential formula using only integer constants, addition, order and equality. Such Presburger-definable sets are semilinear [7, Theorem 1.3]. Terminal slope checks depend only on the fixed schema; they either remove it or leave the same argument.

For a total-event bound E_0 , at most $n + sE_0$ segment identities can have been created. A finite label set and bounded finite list admit finitely many consecutive partitions, legal rule choices and batch assignments with at most E_0 events. Thus there are finitely many schemas. Finite unions of semilinear sets are semilinear. $\square$

More concretely, a semilinear set is a finite union of sets $\mathbf{a} + \sum_{j=1}^m \mathbb{N}\mathbf{b}_j$. The proposition describes a strong boundary: a fixed externally supplied finite schema reduces to integer linear feasibility

despite its quadratic packaging. Strict convexity does not import undecidability into that fixed finite system.

Enumerating all finite schemas gives an enumerable family of quadratics covering finite-event halting, with varying arities and coefficients. In a deterministic machine an actually halting run has a definite finite sequence of complete nonempty batches. That fact does not encode its arbitrary length into one fixed tuple. A single fixed-arity representation of unbounded halting requires an additional history-packing construction, with its own decoder, arithmetic costs and witness-multiplicity proof. Generic Diophantine representability results do not automatically preserve this compiler’s quadratic degree, sparsity, or single witness. Nor does the construction certify continuation through a Zeno accumulation; no finite schema contains infinitely many events.

[write] For a fixed event budget, Part I’s Theorem 10.2 packs the finite union over all first-hit skeletons into one quartic with unique selector and slack witnesses; it does not remove the budget either.

26.1 Strictly positive witnesses

If a convention requires witnesses in $\mathbb{Z}_{>0}$, replace every y_j by $Y_j - 1$ and define

$$Q_S(\mathbf{p}, \mathbf{Y}) = P_S(\mathbf{p}, \mathbf{Y} - \mathbf{1}).$$

This is an integer polynomial of the same degree. The coordinatewise translation is a bijection from positive witness tuples to natural witness tuples and preserves uniqueness and the Hessian. The input-coordinate convention remains unchanged unless it is translated separately and explicitly. This substitution is exact but is not an assertion that conversion is free in an arithmetic-operation ledger.

27 Verification evidence and research questions

The package supplies two exact-arithmetic compilers, a rational event simulator, and deterministic test suites. Arithmetic uses Python integers and `fractions.Fraction`; no floating-point solver or Diophantine oracle is involved. Table 6 summarizes reproduced checks. Counts of certificates include both prefix and halting variants when both are generated; they are not counts of distinct universal-machine programs.

Check	Full layers	Sparse	Additional
Certificates	122	444	878
Exact full-rank tests	30	40	12
Single-coordinate mutations	4 229	10 102	—
One-layer candidate schedules	832	832	—
Terminal certificates	—	198	372
Height checks	—	444	878

Table 6: Finite implementation evidence. The sparse suite includes 1 114 layers, 1 186 events, 3 982 lifetimes and a dedicated hidden-final-collision regression. The additional suite covers 506 fixtures and 2 152 local growth inequalities.

The full-layer suite additionally enumerates all 625 natural tuples in a small witness box for a two-signal example and finds its unique zero. The sparse suite’s original height test uses a

factor-two safety margin; the additional suite checks the sharper constant-one bound (25.1). A release-specific example check verifies the complete annihilation polynomial, the omitted-final-collision obstruction, empty inputs, positive-witness translation, simultaneous events, output fans and the denominator-growth shuttle. Appendix C describes how to reproduce every reported receipt.

These checks can catch coefficient, endpoint and bookkeeping defects. They do not prove an infinite theorem, establish exhaustive novelty, or replace an independent formalization. The mathematical proofs above establish the stated scope.

[write] The counts in Table 6 equal the four expected receipts shipped as `data/11-signal-certificates-expected-*_receipt.json`. The replay was run at the write on a copy with the delivered layout, under Python 3.14.4 on Windows: all four receipts and both coefficient-example files matched. The repository’s review [14] replayed it as well and adds independent finite checks (Section 1.4).

27.1 Questions with concrete next steps

[write] Table 3 pairs these questions with the research directions of Part I: Questions 1 and 5 coincide with Sections 15.2 and 15.1, and Questions 3 and 4 overlap Sections 15.3, 15.4 and 15.5.

1. **Instantiate the universal table and input map.** Transcribe the 13-label, 21-non-blank-rule machine with an explicit convention for every blank collision. Verify the cyclic-tag input encoding and rational halting gadget, then compile and replay complete finite examples. This would connect the present implementation to the cited universal machine rather than only to its speed interface.
2. **Exploit conservative bounded populations carefully.** A reversible conservative rational construction is already Turing universal [5]. Its positive energy assignment bounds the number of simultaneously live signals for fixed total energy, but not the number of collisions. A direct unbounded representation would need to instantiate its finite transition branches, encode unbounded histories of rational distances, and pay for strict guards and input loading. Conservation alone neither bounds runtime nor excludes accumulation.
3. **Sharpen the denominator and cost ledger.** Causal depth can replace batch count in B . Further denominator cancellation may depend on the actual branch structure. A useful next result would make such improvements canonical and give a literal arithmetic-gate ledger, including the construction of coefficients and any conversion of input conventions. ([write] Causal-depth clearing itself is proved: Lemma 19.1 shows that D^H suffices, and Part I, Theorem 7.1, gives D^h event by event. What remains open is a canonical choice compatible with the witness format, and the gate ledger.)
4. **Compress histories without losing canonicity.** The sparse certificate removes repeated geometry, not the unbounded number of events. Can a restricted family of repeated collision motifs be packed with a unique decoder witness and explicit degree, height and operation bounds? Any answer should separate verification of a supplied history from existence of an outcome.
5. **Formalize geometric completeness.** The most valuable proof-assistant target is the typed-endpoint soundness theorem, including atomic simultaneous replacement and disappearing event blocks. The replay data provide regression fixtures, but the formal objects must model the real semantics, not only the executable’s output.

28 Conclusion

A finite signal-machine schema admits a direct, canonical Diophantine encoding because its collision geometry can be expressed by affine integer constraints on a predetermined rational grid. Sparse adjacency lifetimes make the certificate linear in event count and initial support. Correct endpoint types are what turn line-incidence equations into a complete execution certificate. The resulting quadratic is strictly convex in its witness variables, and the four-speed case has an explicit linear witness-bit bound. The remaining unbounded problem is to encode arbitrary schema length without losing the quantitative and multiplicity properties that make this finite construction useful.

A Part I: a minimal worked use of the compiler

The following code can be run from the archive root. ([write] “The archive root” is the delivered layout, with `code/signal_certificates.py`; in this report the module is shipped as `code/07-collision-geometry-signal_certificates.py`, so the snippet runs in a copy that restores the delivered name (README of this report).) It builds the first-collision tournament and constructs a complete certificate without using an external algebra system.

```
import sys
sys.path.insert(0, "code")
from signal_certificates import Machine, compile_skeleton

machine = Machine(
    {"A": 2, "B": 0, "C": -1, "H": 0, "R": 0},
    {
        frozenset(("A", "B")): ("H",),
        frozenset(("B", "C")): ("R",),
        frozenset(("A", "B", "C")): ("R",),
    },
)
skeleton = (((0, 1),),)
cert = compile_skeleton(machine, ("A", "B", "C"), skeleton)
gaps = (1, 1)
slacks = cert.witnesses(gaps)
assert cert.evaluate(gaps, slacks) == 0
assert not cert.accepts((3, 1)) # An earlier BC collision.
assert not cert.accepts((2, 1)) # A simultaneous ABC collision.
print(cert.to_dict())
```

The order of rows in the exported matrices is lexicographic and need not match the order used in the displayed mathematical formula. Row order changes only the order of slack coordinates, not the represented relation or the uniqueness statement.

B Part I: dependency map and claim status

Claim	Principal ingredients	Status in this package
Exact chronology chamber	Intercept induction; affine endpoint lemma; static identity and order validation	Full manuscript proof; exact tests
Event-sparse row bound	Local adjacency replacement count	Full manuscript proof; every exported ledger checked
Unique-slack quadratic	Integer strict inequalities; sum-of-squares zeros	Full manuscript proof; zero and mutation tests
Canonical first-hit quartic	Disjoint first-hit chambers; natural one-hot selector equation	Full manuscript proof; finite union tests
Causal denominator and coefficient bounds	Integer speed differences; induction on causal depth	Full manuscript proof; finite denominator checks
Small integral realization	Homogeneity; rational vertex; Cramer’s rule; Hadamard bound	Full manuscript proof
Robustness and codimension	Linear margin estimate; rank of equalities	Full manuscript proof
Finite-horizon semilinearity	Exact chamber theorem; classical Presburger and semilinear-set equivalence	Derived theorem using cited literature
Exact quasi-polynomial counts	Chamber homogeneity; face inclusion–exclusion; classical Ehrhart theorem	Derived theorem; worked counts checked
Universal signal-machine existence	Durand-Lose’s published construction	External result; rule table not reimplemented
Fixed-arity unbounded Diophantine existence	Computable finite-prefix simulation; classical MRDP	External existence theorem, not new compiler
Global convex-quadratic barrier	Positive semidefinite quadratic completion; Presburger decidability	Full manuscript proof

[write] The row “Global convex-quadratic barrier” is a special case of Part XI of [11] (Remark 13.3); source 07’s proof is kept as a second route. The rows “Exact chronology chamber”, “Event-sparse row bound”, “Unique-slack quadratic”, “Causal denominator and coefficient bounds” and “Finite-horizon semilinearity” have second proofs in Part II (Section 1.2).

C Part II: replay and build guide

The accompanying archive contains this article in editable LaTeX, the compiled PDF, exact Python scripts, expected receipts, example coefficient dictionaries, and a manifest with file hashes. It contains no third-party source PDFs. The scripts require Python 3.10 or later and only the standard library. Assertions must remain enabled; do not run Python with `-O`.

From the extracted package root run:

```
python3 scripts/run_replay.py
```

The runner copies the executable scripts into a temporary directory, executes the suites there, compares fresh JSON receipts to the stored expectations, and writes a summary under `replay-output`. The stored expected receipts and packaged examples are not overwritten. Use `-output DIR` to select a different result directory. The worked-example receipt also contains its concrete witness and explicit rejection reason for the false final-cut schema.

[write] These are the delivered names. In this report the five scripts are shipped as `code/11-signal-certificates-*.py`, `build.sh` as `code/11-signal-certificates-build.sh`, the two example files and the manifest as `data/11-signal-certificates-*.json`, and the four expected receipts as `data/11-signal-certificates-expected-*_receipt.json`. The runner looks for `scripts/`, `examples/` and `expected_receipts/` in the parent of its own directory and runs the scripts by their delivered names, so it runs only in a copy that restores the delivered layout (see the README); it writes nothing into the shipped files. Of the archive contents listed above, the editable source is printed as this Part rather than shipped separately, and the compiled PDF and the checksum file `CHECKSUMS.sha256` are not shipped (the checksum file was verified, 17 of 17 files, at placement); the manifest is shipped, and its payload list still names the delivered paths and the delivered PDF and article source.

The main executable interfaces are:

- `signal_geometry.py`: exact simulator, full-layer compiler, polynomial evaluation and rational Gaussian rank calculation;
- `signal_sparse.py`: event-coordinate and adjacency-lifetime compiler;
- `verify_additional.py`: separately seeded edge cases and sharp height checks;
- `verify_examples.py`: the article’s explicit examples and convention checks.

The low-level compilers use assertions to reject malformed schema partitions or structurally impossible collision directions. They are reference mathematical implementations, not hardened parsers for untrusted inputs. JSON residual dictionaries use `@` for the constant term, `x0`, `x1`, and so on for input positions, and the listed variable names for natural witnesses. A certificate evaluates to the sum of the squares of those rows. No expanded polynomial is required to replay it.

To rebuild the article run:

```
sh build.sh
```

A working TeX Live installation with `pdflatex`, Latin Modern fonts, AMS packages, `geometry`, `microtype`, `booktabs`, `enumitem` and `hyperref` suffices. The build script uses fixed document metadata and a fixed source-date epoch; exact PDF byte equality across different TeX distributions is not promised. The mathematical replay receipts are exact and version-independent within supported Python semantics.

D Part II: claim and source guide

The source-supported context is limited to the following claims. All certificate theorems and proofs in this article are supplied here rather than attributed to the literature.

- Durand-Lose 2009: finite configurations in Definition 1; integer inputs in Section 2; speeds in Figure 4; cyclic-tag simulation and halting in Section 5 and Theorem 5. Common-denominator rescaling is our inference, proved in (19.1).

- Durand-Lose 2007: linear-register event simulation in Section 4 of the author version, especially its ninth PDF page; rational models in Section 5. No Diophantine theorem is attributed to it.
- Durand-Lose 2006: energy and live-population bounds in Definition 3 and the following paragraph; accumulation qualifications in Sections 2.2 and 6.
- Ginsburg and Spanier 1966: equivalence of Presburger-definable and semilinear subsets of $\mathbb{N}^n$, Theorem 1.3, pp. 288–289.

Primary sources were checked on 2 October 2026. The bibliography links to the papers or their official bibliographic records; copyright-bearing PDFs are not redistributed in this package.

[write] In this report the Durand-Lose 2009 and Ginsburg–Spanier entries are merged with Part I’s entries for the same papers [3, 7]; the merged entries add source 11’s editors and links to Part I’s.

E Provenance of this report

[write] The report was built from two manuscripts that arrived together in the archives of commit 808b53ed8 (2 October 2026) and were placed by commit 798b0c5d4, which staged their programs, data and provenance files byte-identical to the delivery.

Source	Archive (batch 78)	Pin	Contributed	Printed as
07 (base)	Collision_Geometry_Diophantine_Signal_Machines.zip, manuscript 07; main file collision_geometry/article.tex, 29-page PDF	f1edb38f9	the chamber theorem, the event-sparse bound, the unique-slack quadratic, denominators and heights, small realizations, the NP bound, robustness, semilinearity, the first-hit quartic, Ehrhart counts, the tournament, the clock, the barrier, nine research directions	Part I with Appendices A and B
11	Signal_Machine_Diophantine_Certificates.zip, manuscript 11; main file signal-diophantine-release/article.tex, 15-page PDF	none named	the same theorem with event-coordinate witnesses, halting semantics, full column rank, the size ledgers, the full-layer construction, the four-speed height bound, the bounded shuttle, positive witnesses, five questions	Part II with Appendices C and D

Source 07’s pin is the full commit f1edb38f93aa0a362a34b531a600909f6e74608a, named in its Section 14 and in its delivered PROVENANCE.md (shipped as 07-collision-geometry-PROVENANCE.md); it is an ancestor of the placement commit, and the audit blob it cites is unchanged at the write. Source 11 names no repository snapshot and cites no repository file; its only repository-related statement, that the 75- and 87-operation constructions are not improved, is true at the write.

Where the merge had to choose.

- *Base and order.* Source 07 is the base (Part I): it allows signed rational speeds without a normalization interface, proves more about the chamber, and is the longer manuscript. Source 11 follows as Part II. Every result, proof, example, remark, research question, appendix and limitation of both sources is printed, in the sources' order and with their numbering inside each Part.
- *The common theorem.* Its two statements have different inputs and witnesses, so both are printed, each with its proof. The lemmas of Part II that coincide with lemmas of Part I (Lemmas 19.1, 20.1 and 21.1, Proposition 26.1) open with a [write] note naming their Part I counterpart, and source 11's proofs are kept as second routes. Section 1.2 compares the two routes, including the two shuttles and the two missed-simultaneity examples, which are kept in both Parts because they are different machines.
- *A result already in the collection.* Part I's Theorem 13.1 and Corollary 13.2 are a special case of Part XI of *canonical-diophantine-certificates*. They are printed with source 07's statements and proofs, as a second route, and credited in Remark 13.3; credit notes were added to source 07's abstract and conclusion. Remark 10.3 relates the first-hit quartic to the same Part.
- *Research questions.* Both agendas are printed in their Parts; Table 3 pairs the coinciding ones, and one [write] note re-scopes the part of Part II's Question 3 that both sources already prove.
- *Front matter and bibliography.* The two title pages are replaced by one; each source's title, author line, abstract and keywords open its Part. Section 1 is new. The two bibliographies are merged: source 11's keys DL09 and GS66 are cited as source 07's `durandlose` and `ginsburgspanier`, whose entries now carry source 11's editors and links; entries [11, 12, 13, 14, 15] were added for the cross-references.
- *Typesetting.* Equations are numbered by section in both Parts (source 07 numbered them consecutively). Source 11's macro `\N` is written `\MN` because source 07's `\N` prints $\mathbb{N}_0$; both print the natural numbers $\{0, 1, 2, \dots\}$ as their sources did. No symbol was renamed in the printed text and no normalization changed.
- *Added text.* Every sentence written in the merge is marked [write]: Section 1; the openings of the two Parts; notes in Part I in Sections 2, 5, 6, 7, 10 (and Remark 10.3), 11, 12, 13 (Remark 13.3), 14 (shipped file names, the pin), 15 and 16, and in its abstract; notes in Part II in Sections 17, 18, 19, 20, 21, 24, 25, 26 and 27; notes in the four source appendices; this appendix; and the bibliography entries marked [write]. Labels were added to the first lemma of Part I's Section 5, to one example subsection, the research directions, the conclusion and the second appendix of Part I, and to several sections, one corollary, the questions, the conclusion and the second appendix of Part II.

Files. The report directory holds this text (`article.tex`, `article.pdf`), a README, source 07's `PROVENANCE.md` and source 11's `REPRODUCIBILITY.md` (as `07-collision-geometry-PROVENANCE.md` and `11-signal-certificates-REPRODUCIBILITY.md`), and the two sources' programs in `code/` and data in `data/`, under the prefixes `07-collision-geometry-` and `11-signal-certificates-`. The README maps each delivered name to its shipped name and gives rerun instructions that do not overwrite the shipped files. Source 11's manuscript, both PDFs, source 11's delivered README and both checksum ledgers (source 07's `SHA256SUMS`, 14 of 14 files, and source 11's `CHECKSUMS.sha256`, 17 of 17, both verified at placement) are not shipped; they survive in the

archives of commit `808b53ed8`. Source 07's delivered README was the placed `README.md` and has been replaced by the report README.

References

- [1] Vladimir Reshetnikov. *ProveIt*, especially `Computability/HilbertTenthProblem/README.md`. Repository inspected 2 October 2026. <https://github.com/VladimirReshetnikov/ProveIt>.
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